

Dying Better: Hospice Care and the End of Life

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Abstract

More than half of older Americans now receive hospice care before death. Using 1,518 county-level hospice agency entry events between 1999 and 2019, we study how expanding hospice availability affects public spending and patient well-being near the end of life. We estimate that increased access to end-of-life hospice care generates a roughly 2-to-1 fiscal return: counties with hospice agency entry experience a \$4.26 increase in hospice spending per Medicare beneficiary-quarter (7.1% relative to baseline) but a \$9.00 reduction in total non-hospice spending, primarily through reduced hospitalizations and post-acute care. Patients spend fewer of their final days in institutions, receive less intensive and more comfort-oriented care, and are more likely to die at home rather than in an institution. We find no adverse effects on pain, functional status, or mortality. Despite these benefits, hospice utilization varies widely across patients. A model of hospice referral and enrollment incorporating patient severity, provider financial incentives, and hospice access constraints suggests that reducing financial disincentives and easing barriers to access could increase patient welfare and generate significant fiscal savings.

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Across aging societies—from Europe to East Asia to the United States—end-of-life (EOL) care spending has become a growing pressure point for policymakers ([OECD, 2024](#)). People are living longer with complex chronic conditions, the returns to late-life medical care are often uncertain, and the choices families confront are deeply personal. Because governments finance most of the care received in the final years of life, these pressures fall directly on public budgets. In the United States, medical care in the last year of life accounts for up to 25% of Medicare spending ([Riley and Lubitz, 2010](#); [French et al., 2017](#)), or over \$250 billion annually—more than the federal budgets of SNAP, TANF, and SSI combined.¹

A substantial share of late-life medical care involves high-intensity, hospital-based treatment. In the final 30 days of life, over half of older Americans are hospitalized and more than one-quarter are treated in an intensive care unit (ICU), despite many reporting that they value comfort and the ability to remain at home ([Gomes et al., 2013](#); [Teno et al., 2013](#)). Hospice care emerged in part to address this disconnect, providing palliative services that emphasize symptom management, psychosocial support, and quality of life rather than further curative treatment, typically in patients’ homes. Its use has expanded rapidly: more than half of elderly decedents now receive hospice services, up from roughly one-quarter in 2000 ([Medicare Payment Advisory Commission 2014](#); [Medicare Payment Advisory Commission 2022](#)).

Despite its promise, the decision to enter hospice care requires a significant trade-off. In Medicare, patients must forgo curative treatment in exchange for palliative support—this may improve comfort, but could also limit access to potentially beneficial care. Furthermore, while hospice is intended to enhance quality of life, its effects on patient well-being remain poorly understood. Unlike outcomes such as service use or mortality, which are readily measured in administrative data, comfort and symptom relief are rarely documented in a systematic way. As a result, there is limited large-scale evidence on how hospice affects the patient experience in the final months of life.

This paper brings new evidence to bear on these questions by studying the impact

¹Medicare total expenditure was 1,122 billion in 2024 ([Centers for Medicare & Medicaid Services, 2025](#)). SNAP: Supplemental Nutrition Assistance Program; TANF: Temporary Assistance for Needy Families; SSI: Supplemental Security Income. SNAP budget was 100.3 billion in 2024, the federal TANF block grant was 16.5 billion, and SSI was 63 billion. ([Congressional Research Service, 2024](#), [U.S. Department of Health and Human Services, 2024](#), [Social Security Administration, 2024](#))

of expanding hospice availability. We construct a stacked difference-in-differences design centered on 1,518 county-level “experiments”, each encompassing a treated county—a county with hospice agency entry—and matched control counties, drawn from the set of ever treated counties. We combine over 20 years of Medicare claims with detailed nursing home assessment records from the Minimum Data Set (MDS), creating a comprehensive panel of older adults across thousands of U.S. counties. Unlike administrative claims—which primarily record billing information and therefore measure utilization, spending, and mortality—the MDS contains detailed clinical assessments completed by nursing-home staff. Linking these assessments to Medicare claims allows us to observe dimensions of patient well-being that have largely been inaccessible in large-scale administrative data, including pain, functional status, cognition, feeding, mobility, and care processes. The MDS covers a substantial share of the hospice population, with over 40% of hospice users residing in a nursing facility.²

We find that hospice agency entry increases hospice spending by \$4.26 per beneficiary-quarter (7.1% relative to baseline). This increase is more than offset by a corresponding reduction in total non-hospice Medicare spending (-\$9.00). That is, for each additional dollar of hospice spending, Medicare saves over two dollars in return. This reduction is driven by reduced inpatient (-\$5.29) and skilled nursing facility (-\$2.12) expenditures, with no statistically significant changes in outpatient, physician, or prescription drug spending.

Beyond cost savings, we document a shift in the nature of end-of-life care among nursing home residents: patients are less likely to receive high-intensity interventions such as catheters (-1.6%) or physical restraints (-4.8%), and more likely to receive comfort-oriented services like speech therapy to help with swallowing (2.7%) and hypnotic-class medications for sleep (4%). Place of death also shifts significantly: for every 1% increase in the share of patients using hospice, the share of patients who die in an institution declines by 0.22% while the share of patients who die at home increases by 0.12%. Importantly, we find no adverse effects on mortality, pain, or functional outcomes.

These results show that expanding hospice availability leads to lower Medicare spending and measurable improvements in end-of-life care. Yet hospice take-up and spending offsets are strikingly uneven across different health conditions. The baseline rate of hospice use varies significantly by diagnosis, as does the first-stage response: Alzheimer’s shows

²Author’s analysis of the MDS and Medicare claims data in 2019.

large gains, while cancer and stroke have more modest responses. The size of spending offsets also differs sharply. For Alzheimer’s and related dementias, each additional dollar of hospice spending roughly breaks even in total Medicare savings, whereas for conditions such as ischemic heart disease or COPD, total non-hospice spending falls by about three dollars for every additional dollar spent on hospice.

Despite this heterogeneity in take-up and fiscal savings, the benefits to patients appear consistent. Measures of patient well-being improve across all major conditions, whereas we find no evidence of adverse mortality effects on any condition. The puzzle is therefore not whether hospice can reduce spending or improve outcomes, but why some patients and conditions take up hospice at higher rates than others following an expansion of hospice availability.

We outline three mechanisms that could generate this variation. First, patients differ in their prognosis and expected survival. When patients are less severely ill or have longer expected survival, continued treatment remains more valuable relative to hospice. Variation in hospice use may therefore reflect efficient targeting based on clinical severity. Second, referring providers face financial incentives that vary by the availability and profitability of curative treatment options. When continued treatment generates greater revenue, providers may delay or avoid hospice referral even when hospice would improve patient well-being. Third, patients may face constraints on hospice access even when a provider would otherwise recommend it. These include both supply-side constraints and patient-specific barriers. The discrete increase in hospice use following agency openings indicates that supply-side availability matters. For example, limited agency staffing can constrain enrollment, and policies such as certificate-of-need laws can worsen entry and competition. Patient-side barriers are also significant; for instance, the absence of a caregiver or a suitable home environment often makes receiving home-based hospice difficult or infeasible.

To quantify the relative importance of these mechanisms and conduct policy counterfactuals, we develop and estimate a model of hospice referral and enrollment. In the model, a provider observes the patient’s prognosis and chooses whether to recommend hospice or continued treatment, weighing patient well-being against the financial returns associated with each option. The resulting decision follows a simple cutoff rule: the provider recommends hospice when its expected benefit to the patient, weighted by the provider’s

concern for patient welfare, exceeds the revenue forgone from referral. Hospice access constraints—which encompass both supply-side capacity limitations and patient-side barriers—then raise the effective enrollment threshold above the provider’s referral threshold, further limiting hospice use. The model therefore decomposes variation in hospice use into differences in clinical severity, provider financial incentives, and constraints on hospice access.

We estimate the model using patient-group-level data—defined by demographic and clinical profiles—on hospice participation, baseline mortality, and our estimated responses of hospice use and provider revenue to increases in hospice capacity. The estimated parameters allow us to quantify the relative importance of each mechanism in shaping hospice use and to simulate how participation, spending, and welfare would change under counterfactual policies that remove financial disincentives or expand access. On average across conditions, we estimate that providers refer to hospice when quarterly mortality exceeds 11.7%, although this cutoff varies between 6.7% and 17.4% across patient groups. We find that these severity cutoffs vary systematically with provider incentives: conditions with greater revenue losses have higher cutoffs for hospice referral. Providers trade off patient severity and revenue losses, but are generally more sensitive to patient severity: each 0.1 percentage point increase in mortality risk offsets \$5,811 in lost revenue. By comparison, the average revenue loss from referring a patient to hospice is \$36,474. Our model estimates, combined with the empirical finding that hospice does not increase mortality, suggest that hospice care is currently underutilized. We show that counterfactual policies expanding hospice capacity and mitigating financial incentives could increase the patient welfare derived from by as much as 64%.

This paper contributes to several related literatures. It contributes most directly to the literature on hospice care and end-of-life care. Empirical work shows that although spending in the last year of life is substantial, it is concentrated among patients whose needs are hard to predict *ex ante*, making targeted cost containment difficult (Einav et al., 2018; Zeltzer et al., 2023). The economics literature on hospice care is relatively sparse. Recent work highlights how supply-side incentives shape hospice providers’ behavior: providers adjust enrollment as they come close to Medicare’s reimbursement cap (Coe and Rosenkranz, 2025), Medicare reimbursement schemes affect provider incentive for quality (Alam, 2025), and ownership structure influences the types of patients hospices serve (Dalton and Brad-

ford, 2019). Most closely related to this paper, Gruber et al. (2025) shows that for-profit hospice entry reduces Medicare spending among patients with Alzheimer’s and related dementias (ADRD) by offsetting more intensive end-of-life care. We extend this literature in three directions. First, whereas Gruber et al. (2025) studies for-profit hospice entry among patients with ADRD, we estimate the effects of hospice agency entry nationally across all ownership and patient types, allowing us to characterize both spending effects and patient outcomes across the broader Medicare population. Second, we substantially broaden the set of patient outcomes that can be studied in administrative health data. The health economics literature has largely relied on claims-based measures such as spending, hospitalization, and mortality because administrative data rarely contain direct information on patients’ clinical condition or quality of care. By linking Medicare claims to the MDS, we are able to measure patient well-being using detailed clinical assessments—including pain, functional status, cognition, feeding, mobility, and care processes—that are typically unavailable in administrative datasets. To our knowledge, this is the first large-scale quasi-experimental study to use these assessments to evaluate hospice care.³ Finally, we develop an empirically grounded model of hospice referral behavior that relates hospice utilization to patient severity, provider incentives, and access constraints. The model helps interpret the substantial heterogeneity in hospice utilization across conditions and evaluates counterfactual policies that could improve patient welfare while reducing public spending.

More broadly, this paper contributes to the economics literatures on health care spending and the efficient allocation of medical care. Prior work shows that supply-side incentives, practice styles, and the use of intensive, low-value care are central drivers of U.S. health care costs (Finkelstein, Gentzkow and Williams, 2016; Molitor, 2018; Cutler et al., 2019; Clemens and Gottlieb, 2014) and that many Medicare-financed services deliver limited marginal benefits at very high costs (Doyle Jr, Graves and Gruber, 2017; Einav, Finkelstein and Mahoney, 2023). We contribute by showing that expanding access to a lower-intensity form of care can both reduce Medicare spending and improve measured dimen-

³The medical literature includes several correlational studies on hospice care (Kelley et al., 2013; Obermeyer et al., 2014; Harrison et al., 2022; Shega et al., 2008; Miller, Lima and Mitchell, 2012), but these comparisons are subject to selection bias, as patients choosing hospice may differ in unobserved ways from those pursuing more intensive treatment. A randomized trial of palliative care for metastatic lung cancer showed improvements in quality of life, mood, and survival (Temel et al., 2010), but this model integrated palliative support with ongoing curative treatment, whereas Medicare requires hospice enrollees to forgo such care. Our study provides national quasi-experimental evidence on hospice’s effects under the current Medicare hospice benefit.

sions of patient well-being. We further show that provider incentives contribute to the underutilization of hospice care despite its apparent fiscal and patient benefits.

Finally, the paper contributes to the literature on the fiscal impacts of government policies. Prior work has shown that a small subset of policies pay for themselves, and that these are overwhelmingly investments in children and young adults that generate future fiscal benefits through higher tax revenues ([Hendren and Sprung-Keyser, 2020](#)). Against this backdrop, hospice expansion provides a rare example of a health care service for the elderly that delivers improvements in patient experience while also reducing net public expenditures.

The rest of the paper proceeds as follows. Section 1 introduces the setting. Section 2 describes our data and sample. Section 3 describes our empirical strategy. Section 4 presents our results. Section 5 introduces and estimates our model. Section 6 concludes.

1 End-of-Life Care and Hospice in the United States

End-of-life care in the United States is delivered across several distinct settings, including hospitals, skilled nursing facilities, and patients' homes, with or without hospice services. These settings differ both in the intensity of care provided and in their underlying treatment goals, ranging from curative, hospital-based care to palliative, home-based care.

Figure 1 illustrates the distribution of end-of-life care settings among Medicare beneficiaries in 1999. As shown in Panel A, a substantial share of patients die in institutional settings: 30% in inpatient facilities and 8% in skilled nursing facilities. In comparison, 19% of patients die with hospice care, while 43% die at home without hospice services. Panel B further illustrates the source of hospice admissions, showing that the vast majority of hospice patients are admitted from home, with relatively few admissions originating from inpatient (4%) or skilled nursing (1%) settings.

Hospice care provides palliative medical services, pain management, and psychosocial support to patients nearing the end of life. Covered services include physician and nursing visits, medications, durable medical equipment (e.g., hospital beds, wheelchairs), symptom management, grief counseling, and 24/7 on-call support ([NHPCO, 2024](#)). In the United States, hospice care is delivered mainly in patients' homes and nursing facilities, and it is financed chiefly through the Medicare Hospice Benefit, authorized by the

Tax Equity and Fiscal Responsibility Act of 1982 and first implemented in 1983 ([Medicare Payment Advisory Commission, 2019](#)). Although daily per diem rates vary by level of care—routine home care, continuous home care, inpatient respite care, and general inpatient care—routine home care accounts for 98.8% of hospice days and receives the lowest payment rate ([Medicare Payment Advisory Commission, 2025](#)).

To be eligible for Medicare hospice coverage, patients must be certified by a physician as terminally ill with a life expectancy of six months or less and must elect to forgo curative treatment for their terminal condition ([Medicare Payment Advisory Commission, 2019](#)). Once enrolled, patients receive care from an interdisciplinary team, including physicians, nurses, social workers, and chaplains, under a fixed daily payment per beneficiary that is intended to cover all related services. Beneficiaries face only minimal cost-sharing, limited to a nominal copayment (no more than \$5) for outpatient prescriptions and 5% coinsurance for inpatient respite care. ([Medicare Payment Advisory Commission, 2019](#))

Over the past two decades, the use of hospice care has expanded dramatically, particularly among patients with non-cancer diagnoses such as Alzheimer’s disease and heart failure ([De Vleminck et al., 2018](#)). This growth has coincided with a surge in the number of Medicare-certified hospice providers, driven largely by for-profit entrants ([Medicare Payment Advisory Commission, 2021](#)). Between 2000 and 2019, the total number of Medicare hospice providers more than doubled (Appendix Figure [A1](#)), with a large share consisting of for-profit hospices. Hospice use also expanded substantially during this period, from under 20% of decedents in 2000 to 50% in 2019 (Appendix Figure [A2](#)).

2 Data and Sample

2.1 Data Sources

We combine administrative Medicare claims and clinical assessment data to evaluate the effects of hospice agency entry on health care utilization and patient well-being. The primary data sources include 100% samples of Medicare inpatient (IP), outpatient (OP), skilled nursing facility (SNF), home health agency (HHA), and hospice (HOS) claims for the years 1999–2019. These are complemented by 20% random samples of the Carrier (physician office visits) and Part D prescription drug event (PDE) files, as well as the

Medicare Beneficiary Summary File (BSF), which provides enrollment and demographic information. To ensure consistency across files, we restrict analyses to beneficiaries who are alive and enrolled in fee-for-service (FFS) Medicare at the start of each quarter-year. For analyses involving drug spending, we additionally require enrollment in a Part D plan.

To measure patient well-being and clinical practices, we use the Minimum Data Set (MDS), a federally mandated clinical assessment completed by nurses for all residents of Medicare- or Medicaid-certified nursing homes. The MDS contains detailed, standardized information on residents' physical health, functional status, cognitive impairment, mood, pain, treatments, and care plans. Assessments are conducted on admission and at regular intervals thereafter, including quarterly and annual assessments, providing repeated measures of clinically assessed patient outcomes that are unavailable in Medicare claims data. Approximately 40% of hospice patients reside in a nursing home and are therefore covered by the MDS.⁴

Hospice agency characteristics, including ownership status and staffing, are obtained from the Medicare Provider of Services (POS) file.

2.2 Sample Construction and Key Variables

We examine a broad set of outcomes across several domains.

Hospice utilization Hospice spending is measured as Medicare hospice expenditures per beneficiary-quarter. We also measure hospice participation, defined by the share of beneficiaries electing hospice in a given quarter.

Non-hospice Medicare Spending For non-hospice Medicare spending, we evaluate component-specific spending per beneficiary-quarter, using data from inpatient, skilled nursing facility, outpatient, carrier (physician office), and home health agency fee-for-service claims files from 1999-2019.

For Part D prescription drug spending, we compute Part D spending per beneficiary-quarter using the Part D event file, available from 2006-2019, for the baseline FFS population.

We conduct subgroup analyses by chronic condition, including Alzheimer's, ADRD, cancer, COPD, stroke, ischemic heart disease, and chronic kidney disease.

⁴Authors' analysis of the MDS data.

Patient Care and Clinical Outcomes For patients residing in a nursing home, we measure their clinical outcomes using assessment data from the Minimum Data Set (MDS). Outcomes include use of feeding tubes, parenteral feeding, indwelling catheters, physical and chemical restraints, dialysis, IV medications, and various therapy services (speech, respiratory, physical, occupational, psychological). Clinical outcomes include pain scores and frequency, ADL functioning, cognitive and mood assessments, psychosocial well-being, and indicators of falls, incontinence, and pressure ulcers. We only include regularly scheduled assessments (conducted on admission, annually, and quarterly) and exclude assessments conducted due to significant changes in health status and those conducted upon discharge. Where necessary, we harmonize survey questions that were modified between the two MDS versions (MDS 2.0, used from 1999-2010 and MDS 3.0, used from 2010-2019). Details on the construction of MDS outcomes are in Appendix Section [A](#).

2.3 Summary Statistics

Hospice Agencies Between 1999 and 2019, the number of Medicare-certified hospice agencies in the United States more than doubled, rising from just over 2,000 providers in 1999 to over 5,000 by 2019 (Appendix Figure [A1](#)). This expansion was widespread but geographically heterogeneous. As shown in Appendix Figure [A3](#), hospice entry occurred across much of the country rather than being concentrated in a small number of regions.

Table [1](#) summarizes the characteristics of hospice agencies in 2019. On average, agencies employed 30 staff members, with incumbents substantially larger than entrants (47 vs. 19 employees). This size differential is consistent across staff categories: incumbent agencies have more registered nurses and licensed practical nurses (3.2 and 16.2 vs. 1.5 and 5.5), home health aides (9.7 vs. 4.4), and other personnel. Entrants in the baseline sample are somewhat larger than the full entrant group. Ownership structures also differ: nearly 60% of incumbents are non-profit, compared to just 16% of entrants and 21% of those in our baseline sample. Finally, co-location patterns vary substantially—86% of entrants and 81% of those in our baseline sample operate as free-standing hospices, compared to only 51% of incumbents, while incumbents are more likely to be located at hospitals or other health care facilities.

Counties Appendix Table A1 reports county-level characteristics for the counties included in our baseline sample and compares them with all U.S. counties. As expected, baseline counties – those experience any entry during our study period – are larger on average, reflecting the fact that hospice entry is more common in larger markets. Otherwise, they are broadly similar in terms of health care infrastructure, demographic characteristics, and the prevalence of major chronic conditions.

Medicare Beneficiaries Table 2 presents summary statistics for Medicare beneficiaries in treatment counties during the year prior to hospice agency entry. Excluding hospice and prescription drugs, which is not covered for all beneficiaries, average quarterly Medicare spending per fee-for-service beneficiary totaled \$1,711.74. Hospice spending averaged \$60.04 per beneficiary-quarter, or about 3.4% of spending when hospice is included. Inpatient spending was the largest component, averaging \$752.95 per quarter, followed by physician services (\$469.40), outpatient care (\$254.69), and skilled nursing facility services (\$128.99). Home health and prescription drug spending averaged \$116.18 and \$441.64 per quarter, respectively. Mortality rates in the sample averaged 11.56 deaths per 1,000 beneficiaries per quarter.

3 Empirical Strategy

Hospice agency entry is not randomly assigned. Agencies are more likely to enter counties with increasing demand for end-of-life care. A simple comparison of counties with and without hospice entry would therefore confound the effect of entry with underlying differences in market trends. Our empirical strategy addresses this concern by comparing counties at the time they experience hospice agency entry with otherwise similar counties that also receive hospice agency entry during the sample period, but at a different point in time. Identification comes from the timing of entry within these matched sets, rather than from cross-sectional differences between entry and non-entry counties.

We construct our empirical design using a stacked difference-in-differences framework centered on 1,518 hospice agency entry events, each defining a unique “experiment.” For each treated county experiencing an agency entry, we identify two control counties matched on pre-entry trends and levels in the number of Medicare beneficiaries, which capture the

size and trajectory of the local Medicare market. To isolate the effect of the entry, we require that both treated and control counties have no hospice agency entries in the 4 quarters prior to the quarter of agency entry in the treatment county. However, we do not restrict entry in the post-period, avoiding selection on future hospice entry, which may itself be affected by the initial agency entry.⁵ Control counties additionally lack agency entry in the quarter of treatment agency entry. Control counties are drawn from the set of ever-treated counties, ensuring that treated counties are compared to markets that also eventually receive hospice entry. We further require that control counties do not share a border with the treated county to reduce potential spillovers. Details on the matching algorithm are provided in Appendix Section B.

The identifying assumption is that, absent hospice agency entry, outcomes in the treated county would have evolved similarly to outcomes in its matched control counties within the same experiment. The stacked event-study specification provides a direct diagnostic for this assumption: if treated counties were already on different trajectories before entry, we would expect to observe differential pre-trends in the event-time coefficients. As we show in the following section, we find no evidence of differential pre-trends across the outcomes we study.

Starting from 1,828 initial entry events for which there is no preceding entry in the same county in the prior year, we require at least one matched control county and impose a balanced panel, restricting to experiments for which we observe outcomes at least one year prior to agency entry and five years after agency entry. These restrictions bring us to our final sample of 1,518 experiments.

We estimate the effects of hospice agency entry using an event study framework implemented in a stacked difference-in-differences design. Our unit of observation is the county-experiment-quarter.

Our baseline specification is as follows:

$$Y_{ect} = \alpha_{ec} + \gamma_{et} + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t = k] \cdot \mathbf{1}[D_{ec} = 1] + \varepsilon_{ect} \quad (1)$$

where Y_{ect} denotes the outcome of interest for experiment e , county c , and event time t

⁵This design implies that the estimated treatment effects represent the average impact of approximately 1.5 hospice entrants over five years, rather than a strict single-entry effect.

(measured in quarters relative to hospice entry); α_{ec} are county-by-experiment fixed effects to account for persistent differences across counties within each experiment; γ_{et} are experiment-by-time fixed effects that flexibly absorb shared shocks across treated and control counties; and the event-time coefficients β_k capture the differential evolution of outcomes in treated counties relative to controls. The omitted category is $k = -1$, the quarter immediately preceding hospice agency entry. Standard errors are clustered at the experiment level.

To summarize the average effect of hospice agency entry over the post-entry period, we also estimate a pooled difference-in-differences specification:

$$Y_{ect} = \alpha_{ec} + \gamma_{et} + \beta \cdot \mathbb{1}[t \geq 0] \cdot \mathbb{1}[D_{ec} = 1] + \varepsilon_{ect}. \quad (2)$$

Here, $\mathbb{1}[t \geq 0]$ is an indicator for quarters after hospice agency entry.

4 Results

4.1 Effects on Health Care Spending

4.1.1 First-Stage Estimates

We begin by documenting the first-stage relationship between hospice agency entry and hospice utilization. Consistent with our identification strategy, the opening of a new hospice agency leads to a statistically significant and economically meaningful increase in hospice use among Medicare beneficiaries. Figure 2 presents event-study estimates of the effect of hospice agency entry on hospice utilization. Panel A shows the share of beneficiaries using hospice care in each county-quarter, which rises by 0.053 percentage points from a baseline of 0.947 percent, corresponding to a 5.6 percent increase (Table 3 Panel A). Panel B reports hospice spending per beneficiary-quarter, which rises by \$4.26 per beneficiary-quarter from a baseline of \$60.0, or 7.1 percent (Table 3 Panel A). In both panels, we observe an immediate increase in hospice use beginning in the quarter of agency entry, with continued growth over the next eight quarters, and no evidence of differential pre-trends between treated and control counties.

To further assess our identification strategy, we re-estimate the first-stage event study

using a more stringent sample that requires no hospice agency entry during the eight quarters preceding treatment, rather than four (Figure A4). This restriction reduces the number of experiments from 1,518 to 1,147 but leaves the results essentially unchanged. If anything, the estimated effects are slightly larger, providing additional evidence that our findings are not driven by differential pre-treatment trends.

First-Stage Heterogeneity by Major Chronic Conditions Appendix Figure A5 and Appendix Table A2 show that the first-stage response is broad-based across major chronic conditions. Baseline hospice spending varies substantially across diagnoses, ranging from less than \$20 per beneficiary-quarter among beneficiaries without chronic conditions to nearly \$500 among those with Alzheimer’s disease. Consequently, absolute increases in hospice spending also differ considerably across conditions. The largest increases occur for Alzheimer’s disease (\$37.40 per beneficiary-quarter) and Alzheimer’s disease and related dementias (\$28.10), while chronic kidney disease, chronic obstructive pulmonary disease (COPD), congestive heart failure, ischemic heart disease, and stroke/transient ischemic attack (TIA) each experience increases of \$7–\$12 per beneficiary-quarter. Expressed relative to baseline, however, most conditions exhibit increases of roughly 6–8 percent, close to the overall estimate of 7.1 percent. Cancer patients experience a somewhat smaller average increase (5.1 percent), although responses vary across cancer types. Overall, hospice agency entry increases hospice utilization across all major chronic conditions, with some heterogeneity in the magnitude of the response.

4.1.2 Effect of Hospice Agency Entry on Medicare Spending

We next examine the impact of hospice agency entry on total non-hospice Medicare spending and its component parts, including inpatient, skilled nursing facility, outpatient, physician service, and home health agency spending. We analyze prescription drug spending in Medicare Part D separately, since Part D coverage only began in 2006 and is not universal among Medicare beneficiaries. Figure 3 reports the event study for total Medicare spending (excluding hospice and prescription drugs) and Figure 4 breaks down the spending by type. The corresponding aggregate difference-in-differences estimates are reported in Table 3.

Total spending We find a \$9.00 decline in total Medicare spending (excluding hospice and prescription drugs), relative to a baseline of \$1,712 per beneficiary-quarter. That is, an additional hospice agency entry is associated with a 0.53% reduction in Medicare spending in that county. For each dollar of additional hospice spending, total Medicare spending (excluding hospice and prescription drugs) declines by $\$9.00/\$4.26 = \$2.11$.

Spending on Inpatient Hospitalizations The largest component of the observed decline in Medicare spending is attributable to reductions in inpatient hospital expenditures. Following hospice agency entry, inpatient spending falls by \$5.29 per beneficiary-quarter, relative to a baseline of \$753, representing a 0.7% decline. This reduction is consistent with the hypothesis that greater availability of hospice services leads to substitution away from high-intensity hospital-based care.

These spending changes are economically meaningful relative to the first-stage increase in hospice spending. Our estimates imply that each additional dollar in hospice spending corresponds to a \$1.24 decline in inpatient spending, more than offsetting the increase in hospice expenditures.

Spending on Skilled Nursing Facility Care We also find meaningful reductions in Medicare spending on skilled nursing facility (SNF) care. SNF spending falls by \$2.12 per beneficiary-quarter ($SE = 0.59$, $p < 0.001$), relative to a baseline of \$129, corresponding to a 1.6% decline. Considered in the context of our first-stage estimates, each additional dollar in hospice spending corresponds to a \$0.50 reduction in SNF spending. Since Medicare requires an inpatient stay prior to admission by a SNF, reduced inpatient use likely contributed to the reduction in SNF spending.

Spending on Outpatient, home health, and physician services We find no significant change in Medicare spending on outpatient, physician services, and home health care spending. The point estimates—0.04, -1.67, and -0.57 per beneficiary-quarter, respectively—are statistically insignificant and economically small relative to the pre-period means of \$255, \$469, and \$116.

Spending on Part D Prescription Drugs Spending on prescription drugs covered under Medicare Part D is not significantly decreased by hospice agency entry. Average

drug spending per beneficiary-quarter is \$442 at baseline and is reduced by a statistically insignificant \$0.75.

Spending Heterogeneity by Major Chronic Conditions Figure A6 and Table A5 show heterogeneity in the effects of hospice agency entry on total spending, stratified by chronic condition subgroup. Across nearly all groups, hospice entry led reductions in spending, though the magnitude and significance of these effects vary. The largest spending declines are observed among patients with Alzheimer’s (\$38.40), Congestive Heart Failure (\$33.0), and stroke/TIA (\$47.60). Patients with COPD, and ischemic heart disease also experience sizable average spending reductions of \$26.50, and \$22.50, respectively. In contrast, no statistically significant spending reduction was detected for patients with chronic kidney disease or for those with cancer.

Relative to the first stage increase in hospice spending, patients with strokes/TIA saw the largest substitution effect: for every additional dollar spent on hospice care, total non-hospice spending declined by $\$47.60/\$9.71=\$4.90$. Patients with ischemic heart disease, COPD, and CHF also show close to 1-for-3 substitution. In contrast, patients with Alzheimer’s and ADRD roughly break even. This pattern may reflect the fact that for Alzheimer’s and ADRD, there are essentially no curative treatments that the patient must forgo in order to receive hospice care.

4.1.3 Heterogeneity by County and Agency Characteristics

While the baseline estimates capture the average effect of hospice agency entry on hospice and other Medicare spending, impacts may differ depending on local market conditions and hospice agency ownership.

Table A3 shows that both the increase in hospice spending and the reduction in non-hospice spending are larger in counties with low pre-entry hospice utilization. This pattern is consistent with hospice entry expanding access in markets where hospice services were previously scarce. By contrast, in counties with already high hospice utilization, an additional entrant likely serves fewer new patients and instead attracts patients from incumbent providers (i.e., “business stealing”), resulting in a smaller increase in overall hospice use and correspondingly smaller effects on non-hospice spending.

Table A4 examines whether the effects of hospice agency entry differ by agency own-

ership. Both for-profit and non-profit entrants increase hospice spending and reduce total Medicare spending, but the magnitudes differ. The increase in hospice spending is larger following for-profit entry (\$5.23 per beneficiary-quarter, or 8.5 percent) than following non-profit entry (\$2.57, or 4.5 percent), suggesting that for-profit agencies expand hospice utilization more aggressively. Total Medicare spending falls following both types of entry, by \$7.25 (0.4 percent) for for-profit entrants and \$12.06 (0.7 percent) for non-profit entrants, implying a larger offset of non-hospice spending per dollar of additional hospice spending. One possible explanation is differences in case mix. For-profit hospices have a slightly higher concentration of Alzheimer’s disease and ADRD patients, among whom we estimate relatively modest spending offsets, and a lower concentration of chronic conditions for which hospice entry generates larger reductions in non-hospice spending (Appendix Figure A9).

4.1.4 Robustness to Cohort-Based Sample Restrictions

A potential concern with the baseline county-quarter analysis is that the composition of beneficiaries residing in a county may change after hospice agency entry. For example, if beneficiaries move into treated counties after entry, or if post-entry county residents differ systematically from pre-entry residents, estimated changes in hospice use or Medicare spending could partly reflect changes in the beneficiary population rather than changes in care patterns among beneficiaries exposed to the entry event. To address this concern, we re-estimate the main specifications using cohort-based sample restrictions that require beneficiaries to have lived in the experiment county before the hospice agency entered.

These cohort specifications produce results that are very similar to the baseline estimates. Figure A7 shows that hospice agency entry continues to generate a clear first-stage increase in hospice use when the sample is restricted to beneficiaries who were living in the county both in the quarter of observation and in at least one pre-treatment quarter. This result indicates that the increase in hospice utilization is not driven by beneficiaries who moved into treated counties only after the entry event.

The spending results are also robust to this restriction. Figure A8 shows that total non-hospice Medicare spending declines in the cohort sample, closely mirroring the baseline pattern. Similarly, Figure A10 shows that the component-specific results are preserved:

the decline in total spending remains concentrated in inpatient and skilled nursing facility spending, while outpatient, physician, and home health spending show little systematic change.

4.2 Effects on Patient Well-being

4.2.1 Effects on Mortality, End-of-Life Care Setting, and Place of Death

Mortality We find no evidence that increased hospice availability affects aggregate all-cause mortality. As shown in Figure 5, there is no statistically significant change in all-cause mortality rates following hospice agency entry. The first row of Table A6 reports the difference-in-differences estimate. The point estimate for all-cause mortality is -0.017 deaths per 1,000 beneficiary-quarters ($SE = 0.023$), allowing us to rule out mortality increases larger than 0.0028 percentage points, or roughly 0.24% of the baseline rate. Since hospice spending increased by 7.1% (Table 3), our estimates allow us to rule out an elasticity of mortality with respect to hospice spending larger than $0.24/7.1 = 0.034$.

We also do not find significant mortality effects in our robustness specification, when restricting to the cohort of beneficiaries who were living in experiment counties in the year prior to the relevant provider entry events (Figure A13).

A natural interpretation is that the patients induced into hospice by agency entry are not, on average, patients for whom forgone curative or institutional care would have generated substantial survival gains. This would be the case if there are diminishing returns to curative care at the end of life, or if providers selectively refer patients to hospice when expected gains from continued curative treatment are limited. In addition, hospice may provide supportive services that partially offset the withdrawal of some forms of curative or institutional care. Under this interpretation, hospice agency entry changes the setting and intensity of care for marginal patients without measurably shortening survival.⁶

⁶While this is our preferred interpretation, there may be other potential explanations for the lack of mortality effects. First, it may be the case that we are underpowered to detect very modest changes in mortality. Second, hospice agency entry may have heterogeneous effects on mortality across patient populations: while some patients who switch from effective curative care to hospice may experience mortality increases, this may be offset by mortality declines among other patients who benefit from receiving increased services. This is consistent with the mixed prior evidence on specific patient populations: Gruber et al. (2025) finds that for-profit hospice use among AD/DRD patients and increased mortality for marginal patients induced into hospice. In contrast, Temel et al. (2010) find that early palliative care for patients with metastatic non-small-cell lung cancer increased median survival while reducing aggressive end-of-life care.⁷ In our setting, we do not find statistically significant effects on mortality for any chronic condition

End-of-Life Care Setting Although we find no statistically significant change in mortality, we do find meaningful changes in where patients spend their last days of life. Table A7 shows that patients spend a smaller share of their last 6 months of life in an institution (-0.001, a statistically significant 1.1% reduction relative to baseline), driven by reductions of days in both inpatient (-0.0004, or -0.6% relative to baseline) and skilled nursing facilities (-0.0009, or -1.6% relative to baseline).

Place of Death Table 5 reports the estimated effects of hospice agency entry on the place of death. The outcomes are measured as the share of beneficiary deaths that occur in each location. Deaths in an institution decline by 0.4 percentage points (-1.2%), driven primarily by fewer deaths in inpatient facilities (-0.3 percentage points, -1.2%) and, to a lesser extent, skilled nursing facilities (-0.1 percentage point, -1.7%). Deaths at home increase modestly overall (0.4 percentage points, 0.7%), reflecting a substantial rise in deaths at home with hospice care (1.0 percentage point, 3.1%) and a corresponding decline in deaths at home without hospice care (-0.5 percentage points, -1.7%).⁸

Figure 6 further reports these estimates as elasticities relative to the first stage (Table 3), which suggests that a 1.0% increase in the share of beneficiaries using hospice in a calendar quarter corresponds to a 0.55% increase in the share of beneficiary deaths that occur at home with the use of hospice care.⁹

Deaths occurring at home and lower rates of hospitalization near death are commonly used metrics of goal-concordant or high-quality end-of-life care. The significant reductions in institutional deaths and the corresponding increase in home deaths with hospice therefore indicate that hospice agency entry produced measurable changes along these quality dimensions.

4.2.2 Effect on Interventions and Patient Comfort

The claims-based results so far show that hospice entry reduces institutional spending and shifts the place of death. We next ask whether expanded hospice availability also

subgroup (Table A6), although we admittedly have limited statistical power in our subgroup analysis.

⁸Deaths at home include all deaths occurring outside of inpatient facilities or skilled nursing facilities, which may include hospice facilities, nursing homes, or community-based homes. Deaths in inpatient or SNF care are identified as cases where a beneficiary's death date coincides with their discharge date from an inpatient or SNF claim.

⁹Calculated as $0.031/0.056 = 0.554$.

changes the nature of care delivered and patient comfort, focusing on nursing-home residents, for whom we observe rich care processes and outcome measures. We organize these measures into three domains. First, intervention-intensity measures capture invasive or restrictive practices that may be clinically appropriate in some cases but can be burdensome near the end of life. We focus on three common but controversial practices: feeding tubes, indwelling urinary catheters, and physical restraints. Second, comfort-oriented services that focus on symptom management and patient comfort. We focus on swallowing-related speech therapy to promote eating, and hypnotic medications for sleep. Third, physical and mental health measures allow us to assess whether changes in care processes are accompanied by measurable deterioration in patient condition, including pain, mood, cognition, activities of daily living (ADLs), among others.

We start by verifying that hospice agency entry increases hospice utilization among the subset of beneficiaries residing in nursing homes, the population observed in the MDS.¹⁰ Appendix Figure A11 shows that hospice spending and hospice participation increase substantially in this sample as well, although the magnitudes are larger given the much higher baseline use of hospice among nursing home residents. We also confirm that the effects of hospice agency entry on hospice and non-hospice spending are similar among nursing home beneficiaries and the broader beneficiary population (Table A8). Reassuringly, we do not find any evidence that hospice agency entry affects the number of beneficiaries represented in the MDS.¹¹

Intervention Intensity We examine how increased hospice availability affects the intensity and nature of care provided to Medicare beneficiaries residing in nursing homes. We analyze the use of clinical interventions that reflect more aggressive care goals as opposed to prioritizing patient comfort. The most prominent examples are feeding tubes, indwelling urinary catheters, and trunk restraints: tubes can maintain caloric intake and

¹⁰Note that our analysis focuses on long-term nursing home residents, not patients receiving skilled nursing facility (SNF) services. A nursing home is a place of residence that is generally not covered by Medicare, whereas SNF care is a Medicare-covered post-acute service provided following discharge from an inpatient facility. To ensure that our measures are not mechanically affected by hospice enrollment, we restrict the analysis to regularly scheduled MDS assessments, excluding assessments triggered by significant changes in clinical status or by discharge. We also verify that hospice entry does not affect nursing home residence, ensuring that changes in MDS outcomes are not driven by compositional changes in the resident population.

¹¹Point estimate -1.8 beneficiaries (SE = 3.2, P = 0.58) relative to a pre-period mean of 1,077.

lower choking risk, and catheters can prevent urinary retention, yet both introduce substantial discomfort and infection risk (Estrada, Malani and Vitale, 2025). Trunk restraints, such as belts or vests that restrict movement of the upper body, are used in nursing homes to prevent falls or maintain safety for residents with cognitive impairment or physical instability. However, their use is controversial, as restraints can cause discomfort, agitation, loss of dignity, and even physical harm.

Feeding Tubes: Following hospice agency entry, we find a visible decline in the probability that a resident has a feeding tube in place (Figure 7, Panel A). The baseline prevalence of feeding tube use is 7%, and we estimate a reduction of -.09 percentage points. While the aggregate effect across all follow-up periods is not significant, later periods consistently show stronger effects, suggesting a growing impact over time as agency penetration deepens.

Urinary Catheters: We find a modest decline in the share of nursing home residents with an indwelling urinary catheters following hospice agency entry. Use declines by -0.13 percentage points from a baseline of 8%, a 1.6% decrease (Figure 7, Panel B). Importantly, we observe no significant change in urinary continence (Appendix Table A9) or toileting scores (Appendix Table A10), suggesting that this decline does not reflect a shift in patient needs but rather a change in care practices aligned with hospice goals.

Trunk Restraints: We find evidence that hospice agency entry leads to a reduction in the use of trunk restraints in nursing home settings (Figure 7, Panel C). Trunk restraints, which include belts or vests that limit torso movement, are used in 6.37% of resident-quarters at baseline. Following hospice entry, we estimate a decline of 0.29 percentage points (SE = 0.11), a 4.8% decrease. Notably, the reduction in trunk restraint did not result in an increase in falls (Figure A9).

Table 4, Panel A reports the corresponding difference-in-differences estimates.

Comfort-Oriented Services Hospice and palliative care philosophies encourage symptom-directed measures.

Speech Therapy (Swallowing): Consistent with attempts to allow patients to feed themselves, we find a statistically significant increase in the use of speech therapy, which is a service provided to assess and rehabilitate independent swallowing function. That is, patients are receiving support to eat on their own instead of getting calories from a feeding

tube. We estimate an increase of 0.42 minutes per resident-quarter, from a baseline of 15.6 minutes, an increase of 2.7% (Figure 7, Panel D, Table 4). We find no statistically significant changes in respiratory, physical, occupational, or psychological therapy (A11).

Hypnotic Medication: Medications such as hypnotics may help patients sleep better and reduce agitation. We observe an increase in the use of hypnotic medications following hospice agency entry. The probability of receiving any hypnotic medication during a nursing home stay increases by 0.21 percentage points (baseline = 0.06, $p = 0.005$), a 3.5% increase, and the number of days on hypnotics increases by 0.014 days per resident-quarter (baseline = 0.34 days, $p = 0.001$), a 4.1% increase (Figure 7(e-f), Table 4). This may reflect greater emphasis on symptom relief for patients experiencing agitation or insomnia.

In contrast, we find no statistically significant changes in the use of antidepressants, antipsychotics, anxiolytics, or intravenous medications (Table A12). Both the extensive and intensive margins of use remain stable across all specifications, with point estimates near zero. These null results indicate that increased hospice use does not systematically alter psychotropic medication regimens outside the specific domain of hypnotic agents.

Physical and Mental Status Table A9 shows results for physical health outcomes.

We observe no effects on the incidence of falls, pressure ulcers, or incontinence. While the share of individuals gaining weight declined (absolute change -0.00171, baseline 6.4%, $p = 0.013$), which is consistent with the reduction in feeding tube use, there was no significant effect on the share of patients losing weight.

Functional outcomes related to activities of daily living (ADLs) are generally unchanged, with one statistically significant increase in the share of patients requiring support with eating, consistent with our finding of reduced feeding tube use (Table A10). No ADL self-performance measures showed statistically significant changes.

Finally, as shown in Table A13, we detect no significant changes in pain measures, including frequency and intensity. Mood scores and memory function also remain unchanged.

Taken together, these findings suggest that while hospice entry affects specific interventions and treatment intensity, it does not lead to systematic changes in underlying patient condition.

(Lack of) Heterogeneity by Major Chronic Conditions In contrast with the heterogeneity we see in spending outcomes, we find very little heterogeneity across conditions in change in intervention and patient comfort. Figure A12 shows that for each of our key measures of intervention intensity and comfort-oriented services (column), the point estimates across conditions (rows) is very similar. In other words, patients of all major conditions appear to benefit similarly from hospice services.

5 A Model of Hospice Use

Our reduced-form analysis shows that hospice agency entry substantially reduces Medicare spending and improves the quality of end-of-life care. However, these gains are not uniform across conditions. Both baseline hospice use and the first-stage response to agency entry vary considerably: some conditions experience large increases in hospice enrollment, while others respond little or not at all. A key question is whether this heterogeneity reflects efficient allocation based on patient prognosis or inefficiencies arising from provider incentives and access constraints.

First, differences in hospice use may reflect clinical heterogeneity. Patients with some conditions have longer expected survival or greater potential to benefit from curative treatment, making hospice less appropriate. In this case, cross-condition variation would largely represent efficient targeting of hospice care. Second, providers face condition-specific financial incentives. If referring certain patients to hospice entails greater forgone revenue, they may delay or avoid referral even when hospice would improve patient well-being, leading to inefficient underuse. Third, patients may face constraints on hospice access even when a provider would otherwise recommend it. On the supply side, limited agency availability, staffing shortages, regulatory restrictions, and hospice admission incentives may constrain enrollment. The discrete increase in hospice participation following agency entry suggests that such supply-side constraints matter.¹² On the patient side, home-based hospice may

¹²Several institutional features may limit hospice capacity in practice. First, regulatory barriers can restrict entry of new providers. Certificate-of-need (CON) laws remain in place for hospice providers in at least 12 U.S. states and require regulatory approval before new agencies can enter or expand ([National Academy for State Health Policy, 2025](#)). Second, hospice care requires an interdisciplinary team including providers, nurses, social workers, and chaplains, and shortages of health care workers – which are present in many geographic areas – may limit the number of patients that providers can serve ([Health Resources and Services Administration, 2023](#)). Third, Medicare payment rules may discourage enrollment of certain patients. For instance, hospice providers are subject to annual per-patient reimbursement caps, which

be difficult or infeasible without an available caregiver or a safe and stable home environment.

To quantify the relative importance of these mechanisms, we develop a model in which patient severity and provider financial incentives determine the referral threshold, while hospice access constraints determine whether referred patients enroll. The model decomposes cross-group variation in hospice use into differences in clinical severity, provider financial incentives, and access constraints. It also provides a framework for estimating how hospice participation and patient welfare would change if financial disincentives or access constraints were removed.

5.1 Model Setup

Each patient is characterized by clinical severity $s \in S \subset \mathbb{R}_+$, where higher s implies a worse prognosis under continued curative care. Patients can receive either hospice (H) or continued curative treatment (C). Providers observe s and make referral recommendations, while patients follow those recommendations.¹³

Patient welfare Let

$$\Delta U(s) \equiv U(H \mid s) - U(C \mid s),$$

denote the patient’s net benefit from hospice. We assume $\Delta U(s)$ is increasing in severity ($\partial_s \Delta U(s) > 0$), reflecting the declining value of curative treatment as prognosis worsens.

Provider decision Following [Ellis and McGuire \(1986\)](#), providers value both patient welfare and revenues:

$$V(d \mid s) = wU(d \mid s) + R_d,$$

where $w > 0$ captures provider altruism and R_d is the financial return from treatment d .

Let $\Delta R \equiv R_C - R_H \geq 0$ denote the revenue loss from referring a patient to hospice. The may make medically complex patients with long expected hospice stays financially unattractive to hospice providers even when such service would reduce overall medical spending ([Medicare Payment Advisory Commission, 2022](#)). These constraints appear to bind in practice: recent news coverage describes hospice providers turning away eligible patients due to staffing shortages and financial pressures related to Medicare reimbursement rules (e.g., [Graham \(2022\)](#); [Span \(2021\)](#)).

¹³Since hospice requires provider referral, it is generally not possible for patients to enroll against their provider’s recommendation.

provider’s incremental value of hospice, relative to curative care, is

$$\Delta V(s) = V(H \mid s) - V(C \mid s) = w\Delta U(s) - \Delta R.$$

Hospice is recommended whenever

$$\Delta V(s) \geq 0. \tag{3}$$

Since $\Delta U(s)$ is increasing in s , there is a unique referral threshold s^* , where patients with $s \geq s^*$ are referred to hospice.

Let $g(s)$ denote the density of severity. Without access constraints, hospice enrollment is

$$D(s^*) = \int_{s^*}^{\infty} g(s) ds.$$

When hospice capacity is limited, the effective referral threshold increases to

$$\tilde{s} = s^* + \kappa,$$

where $\kappa \geq 0$ summarizes barriers to hospice access, including regulatory restrictions, staffing shortages, and reimbursement rules.

The model highlights two distortions. Larger revenue losses (ΔR) discourage hospice referrals even when hospice maximizes patient welfare, while access constraints ($\kappa > 0$) restrict access by reserving hospice for the most severe patients.

5.2 Parametrization

To take the model to the data, we partition patients into 18 clinically meaningful patient groups, indexed by c , based on demographics and chronic conditions as described in Section C.1 (see Appendix Table A14). Within each group, severity s follows a log-normal distribution,

$$\ln s \sim \mathcal{N}(\mu_c, \sigma_c^2),$$

where s represents each beneficiary’s latent mortality risk, constructed as the quarterly mortality rate per 1,000 beneficiaries.

We assume the net benefit of hospice is linear in severity,

$$\Delta U(s) = U_0 + \lambda s.$$

Substituting into Equation (3) yields the referral rule

$$\beta_0 + \omega s - \delta_c \geq 0,$$

where $\beta_0 \equiv wU_0$, $\omega \equiv w\lambda$, and δ_c denotes the referring provider's revenue loss from referring a beneficiary in group c to hospice. We estimate δ_c directly as the negative ratio of the reduced-form effect of hospice entry on non-hospice spending to its first-stage effect on hospice participation.¹⁴

Absent access constraints, providers refer beneficiaries with

$$s \geq s_c^* = \frac{\delta_c - \beta_0}{\omega}.$$

Allowing for group-specific access constraints, the effective referral threshold becomes

$$\tilde{s}_c = \frac{\delta_c - \beta_0}{\omega} + \kappa_c, \tag{4}$$

where $\kappa_c \geq 0$ captures access limitations.

Baseline hospice use for group c , measured as the share of a group's beneficiaries using hospice in a given quarter, is therefore

$$D_c = 1 - F(\tilde{s}_c; \mu_c, \sigma_c^2),$$

where F is the log-normal cumulative distribution function. Given (μ_c, σ_c^2) , this equation identifies $\tilde{s}_c$.

¹⁴Under this definition of δ_c , the referring provider considers all non-hospice spending. We use this definition for empirical tractability; however, future work could consider provider profits net of costs, which are not observed in our data. Changes in profits may also differ across clinicians of different specialties.

5.3 Estimation

Estimation proceeds in two steps (see Appendix Section C for details). First, we estimate the group-specific parameters $(\tilde{s}_c, \mu_c, \sigma_c^2)$ by matching observed hospice use, mortality rates, and stacked event-study estimates of the effect of hospice entry on hospice participation, non-hospice spending, and mortality among hospice users and non-users. Second, we estimate the common referring provider preference parameters (β_0, ω) by GMM using moment equalities and inequalities, recovering the group-specific capacity wedges κ_c from Equation (4).

Identification Cross-condition variation in hospice use, mortality, and the first-stage response to hospice entry identifies $(\tilde{s}_c, \mu_c, \sigma_c^2)$. Combined with observed referring provider financial incentives δ_c , these moments identify (β_0, ω) . Intuitively, conditions with larger revenue losses from hospice referral require higher patient severity for referral, with the relationship between δ_c and $\tilde{s}_c$ identifying ω .

Because low hospice use may reflect either a low baseline value of hospice (β_0) or binding access constraints (κ_c), these parameters are not separately identified. We therefore normalize $\kappa_c = 0$ for the least access-constrained groups. As a result, our counterfactuals quantify the effects of relaxing access constraints relative to these benchmark groups.

5.4 Model Results

Parameter Estimates Panel A of Table 6 reports the estimated group-specific parameters. Mean severity varies substantially across groups, with quarterly mortality rates ranging from 0.3% to 7.7%. Baseline hospice use likewise ranges from 0.3% to 6.2%, and the first-stage effect of hospice entry varies from 0.01 to 0.34 percentage points, indicating substantial heterogeneity in responsiveness to hospice expansion. Referring provider revenue losses from hospice referral average \$36,474 per patient but vary widely across conditions.

Estimated hospice referral thresholds also differ markedly across groups. On average, patients enroll in hospice when their quarterly mortality exceeds 11.7%, with thresholds ranging from 6.7% to 17.4%. Estimated access constraints are positive for nearly all groups, implying that hospice availability is generally limited and referring providers ration access by prioritizing the sickest patients.

The estimated referring provider preference parameters are economically meaningful. We estimate $\omega = 5.8$ (SE = 1.7), implying that higher patient severity substantially increases the value providers place on hospice referral. The estimate of $\beta_0 = -411$ (SE = 146) implies that, absent access constraints, hospice becomes the preferred option once quarterly mortality reaches roughly 7.1%.

Drivers of Hospice Use Figure 8 illustrates the determinants of cross-condition variation in hospice use. Panel (a) shows that baseline hospice use is strongly positively correlated with mean severity, indicating that much of the observed variation reflects differences in patient prognosis.

Panel (b) shows that conditions with larger revenue losses tend to have higher hospice referral thresholds, although the relationship is not statistically significant. Revenue losses are uncorrelated with mean severity (correlation coefficient -0.17, $P = 0.50$), suggesting that referring provider financial incentives are largely orthogonal to underlying clinical need.

Panel (c) shows a strong positive relationship between referral thresholds and estimated access constraints. Access constraints are also strongly positively correlated with mean severity (correlation coefficient 0.73), implying that access is most constrained for the sickest patient groups.

5.5 Counterfactuals

We consider four counterfactuals: (i) the observed equilibrium; (ii) eliminating referring provider financial incentives by setting $\delta_c = 0$; (iii) eliminating access constraints by setting $\kappa_c = 0$; and (iv) eliminating both distortions.

Table 7 shows that access constraints play a substantially larger role than referring provider financial incentives in limiting hospice use. Baseline hospice participation averages 1.8%. Removing financial incentives increases participation modestly to 2.0%, whereas removing access constraints raises participation to 6.9%. Eliminating both distortions further increases participation to 7.5%. These increases reflect progressively lower hospice referral thresholds, with the common threshold under the final counterfactual equal to the point at which hospice first generates positive patient utility.¹⁵

¹⁵Without access constraints or financial incentives, all groups share the same referral threshold. Differ-

The expansion of hospice lowers the average utility of hospice users, as newly admitted patients are less severely ill, but substantially increases aggregate patient welfare. Relative to the baseline, eliminating both distortions increases total hospice-derived utility by 63.6%, with most of the gain attributable to relaxing access constraints.

The model also predicts sizable Medicare savings. Eliminating financial incentives reduces annual Medicare spending by \$4.9 billion, eliminating access constraints saves \$39.5 billion, and removing both saves \$53.6 billion (Appendix Table A15, Appendix Figure A15). Consistent with the reduced-form evidence, the largest savings occur for beneficiaries with cardiovascular, pulmonary, and kidney disease, whereas patients with Alzheimer’s disease and related dementias experience large increases in hospice use but relatively modest—or even slightly negative—spending effects (Figure 9).

Discussion The large expansion in hospice use implied by the model is striking, but it is consistent with our reduced-form evidence that hospice has few detectable downsides for the marginal patient, particularly when little curative treatment remains to be foregone. Under the preferred specification, 7.5% of beneficiaries would derive positive utility from hospice. In practice, however, Medicare requires physicians to certify a life expectancy of six months or less, and reimbursement rules penalize hospice agencies caring for long-stay hospice patients. These institutional features likely account for an important share of the access constraints identified by the model.

There are two important limitations to keep in mind when interpreting our results. First, our measure of referring provider financial incentives, δ_c , is best interpreted as a proxy measure of financial stakes and may not fully capture all financial incentives faced by individual providers or the health systems in which they practice. Moreover, δ_c may partly reflect patient preferences that providers incorporate into referral decisions rather than financial incentives alone. Providers may also differ in altruism (ω) and in the components of spending they internalize. Second, our welfare analysis is partial equilibrium. A complete social welfare analysis would also incorporate Medicare’s cost of providing hospice care and the fiscal cost of offsetting providers’ forgone revenue. These costs are likely to rise as hospice expands to lower-severity patients. Therefore, our estimated savings likely represent an upper bound.

ences in hospice participation arise solely from differences in the severity distribution.

6 Conclusion

In this paper, we find that expanding hospice availability lowers end-of-life spending while enhancing patient well-being at the end-of-life, with no adverse impacts on mortality, physical functioning, or mental wellbeing. On the margin, each dollar spent on hospice reduces other Medicare spending by just over two dollars. The magnitude of these effects varies by condition: spending offsets are largest for stroke, COPD, and heart disease, and smaller for AD/ABRD and cancer. To explain this heterogeneity and to predict the effects of expanding hospice access on patient welfare and spending, we develop a model in which providers balance patient benefits and financial incentives, subject to access constraints. We find that these distortions—financial disincentives and supply constraints—together contribute to the underuse of hospice relative to the social optimum.

Our findings leave open several avenues for future research. First, the estimated effects reflect hospice expansions over a two-decade period during which the industry shifted from predominantly nonprofit to largely for-profit ownership. Future work could examine how organizational form and entry composition mediate the welfare gains from hospice growth, and whether competitive dynamics among providers alter the efficiency–quality trade-off. Second, our estimates identify the response of marginal patients whose hospice use increases with local capacity; effects on inframarginal patients may differ. Third, while our quality measures draw on detailed nursing home assessments, extending the analysis to home-based settings would provide a more comprehensive view of comfort and care quality near the end of life. Finally, our model emphasizes provider decision-making under financial and access constraints; however, our data do not allow us to discern the degree to which hospice participation decisions are affected by patient, family, and physician preferences. Distinguishing between these components of hospice decisions would yield richer insights into how demand- and supply-side factors jointly determine hospice utilization and welfare.

Taken together, our findings suggest that on the margin, the U.S. health care system devotes too few resources to palliative care relative to its social value. Policies that expand hospice access and reduce financial disincentives for hospice referral could simultaneously improve quality of life at the end of life and reduce overall Medicare spending. In practice, expanding hospice availability may involve reducing regulatory barriers to provider

entry or addressing shortages in the interdisciplinary workforce required to deliver hospice services, while aligning payment incentives could reduce the financial losses providers face when referring patients to hospice. More broadly, these results illustrate how expanding availability of lower-intensity end-of-life care options can yield both fiscal savings and welfare gains, by allocating resources toward services that avert costly institutional care and make life's final days more comfortable.

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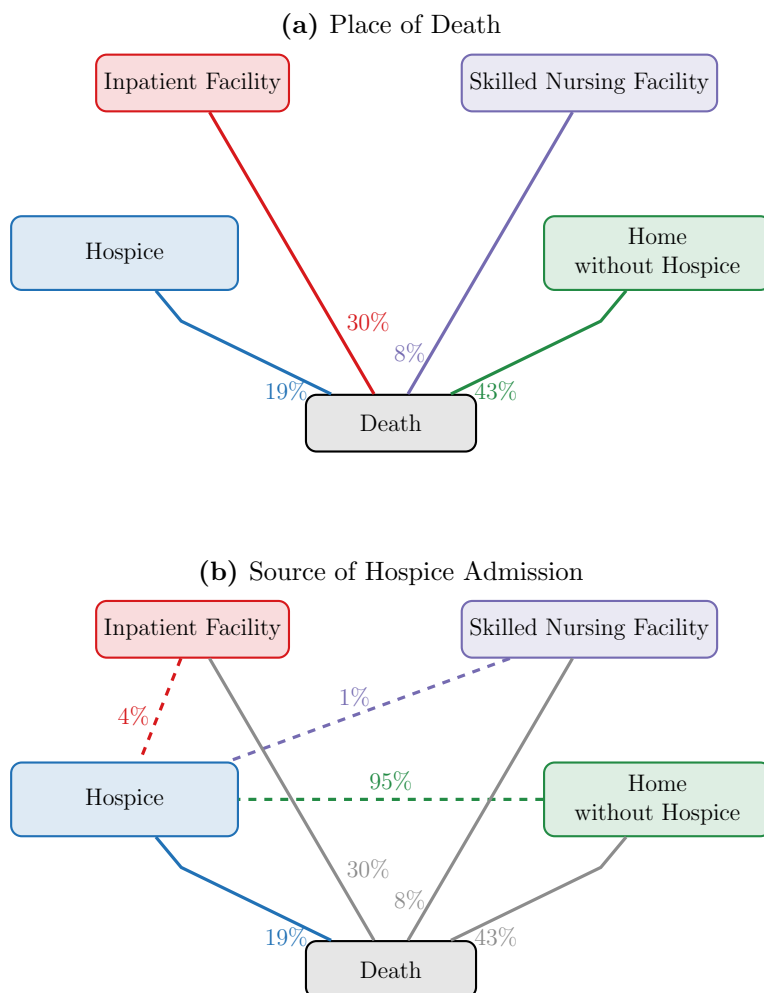
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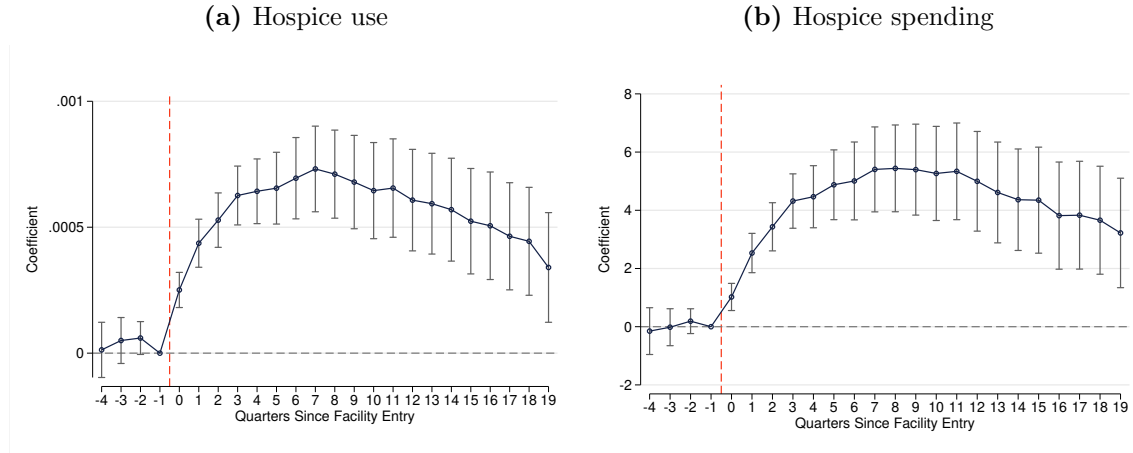
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Figure 1. End-of-life Care Trajectories, 1999



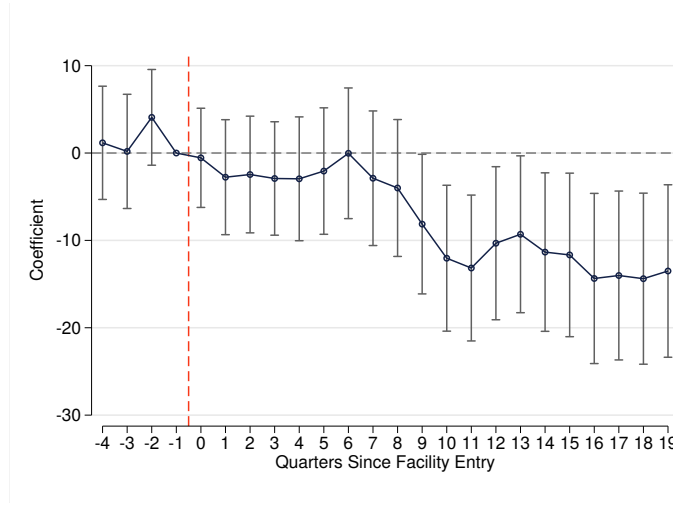
Note: Panel A shows the distribution of deaths across inpatient facilities, skilled nursing facilities, hospice, and home without hospice among Medicare decedents in 1999. Panel B further shows the composition of hospice deaths by prior care setting.

Figure 2. Event Study: Hospice Utilization



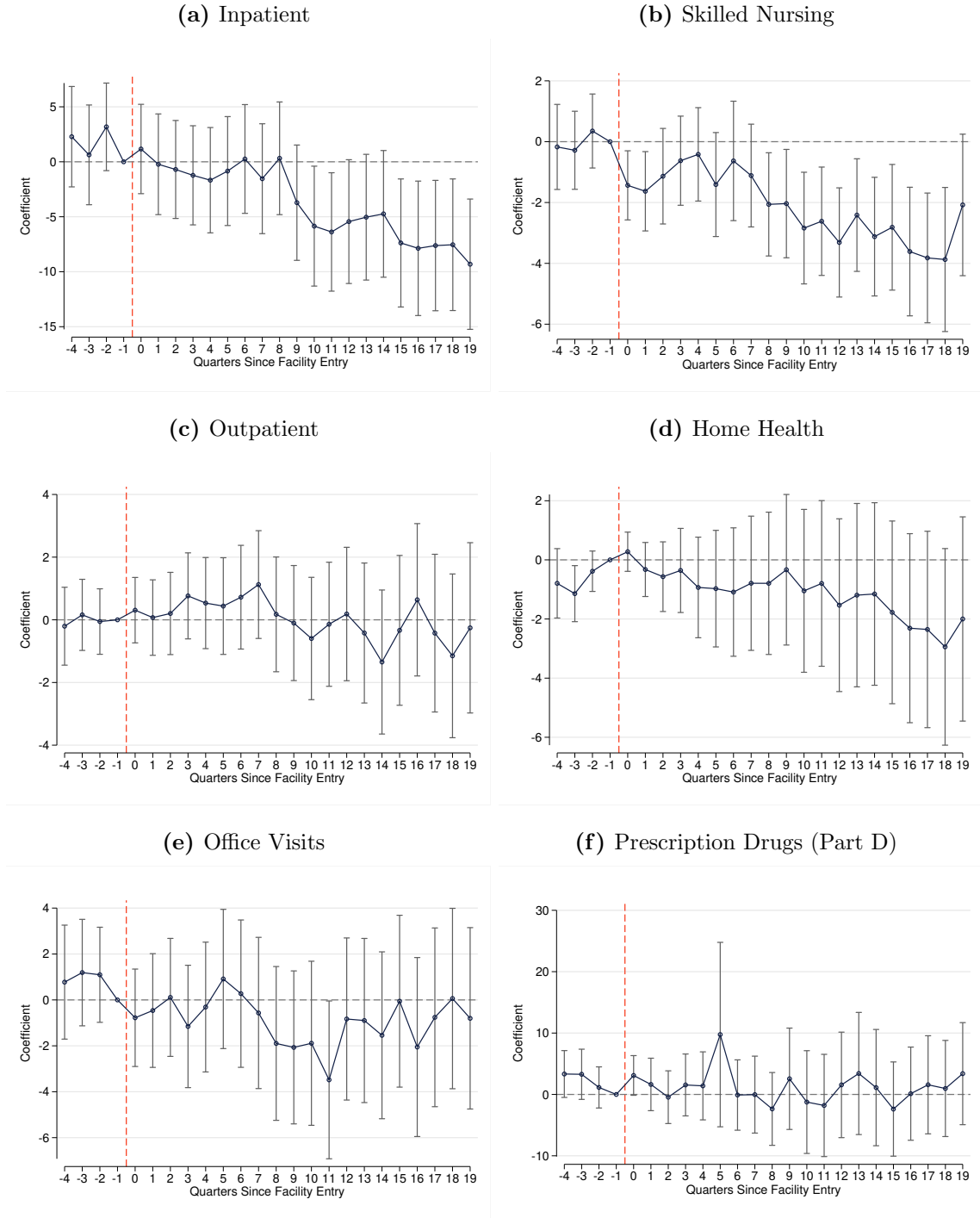
Note: This figure presents estimates from Equation (1) of hospice agency entry on the share of beneficiaries who use any hospice in Panel A and on Medicare hospice spending per beneficiary per quarter in Panel B. Estimates are based on a stacked difference-in-differences event study design using 1,518 hospice agency entries, each defining a separate experiment with one treated county and up to two matched control counties selected based on pre-entry Medicare beneficiary levels and trends. The outcome is measured at the county-experiment-quarter level and modeled with county-by-experiment and experiment-by-time fixed effects. Event-time coefficients represent differences between treated and control counties, normalized to zero in the quarter prior to entry ($k = -1$). The red dashed line marks the timing of facility entry. Standard errors are clustered at the experiment level and shown as 95% confidence intervals.

Figure 3. Event Study: Total Medicare Spending, Excluding Part D and Hospice



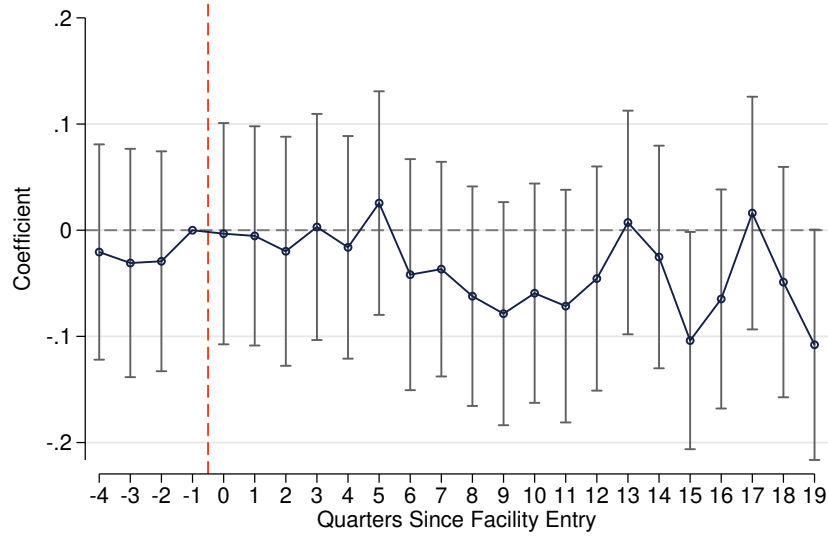
Note: This figure presents estimates from Equation (1) of hospice agency entry on total quarterly Medicare fee-for-service spending per beneficiary, excluding spending on hospice and Part D. Estimates are based on a stacked difference-in-differences event study design using 1,518 hospice agency entries, each defining a separate experiment with one treated county and up to two matched control counties selected based on pre-entry Medicare beneficiary levels and trends. The outcome is measured at the county-experiment-quarter level and modeled with county-by-experiment and experiment-by-time fixed effects. Event-time coefficients represent differences between treated and control counties, normalized to zero in the quarter prior to entry ($k = -1$). The red dashed line marks the timing of facility entry. Standard errors are clustered at the experiment level and shown as 95% confidence intervals.

Figure 4. Event Study: Medicare Spending, by Type



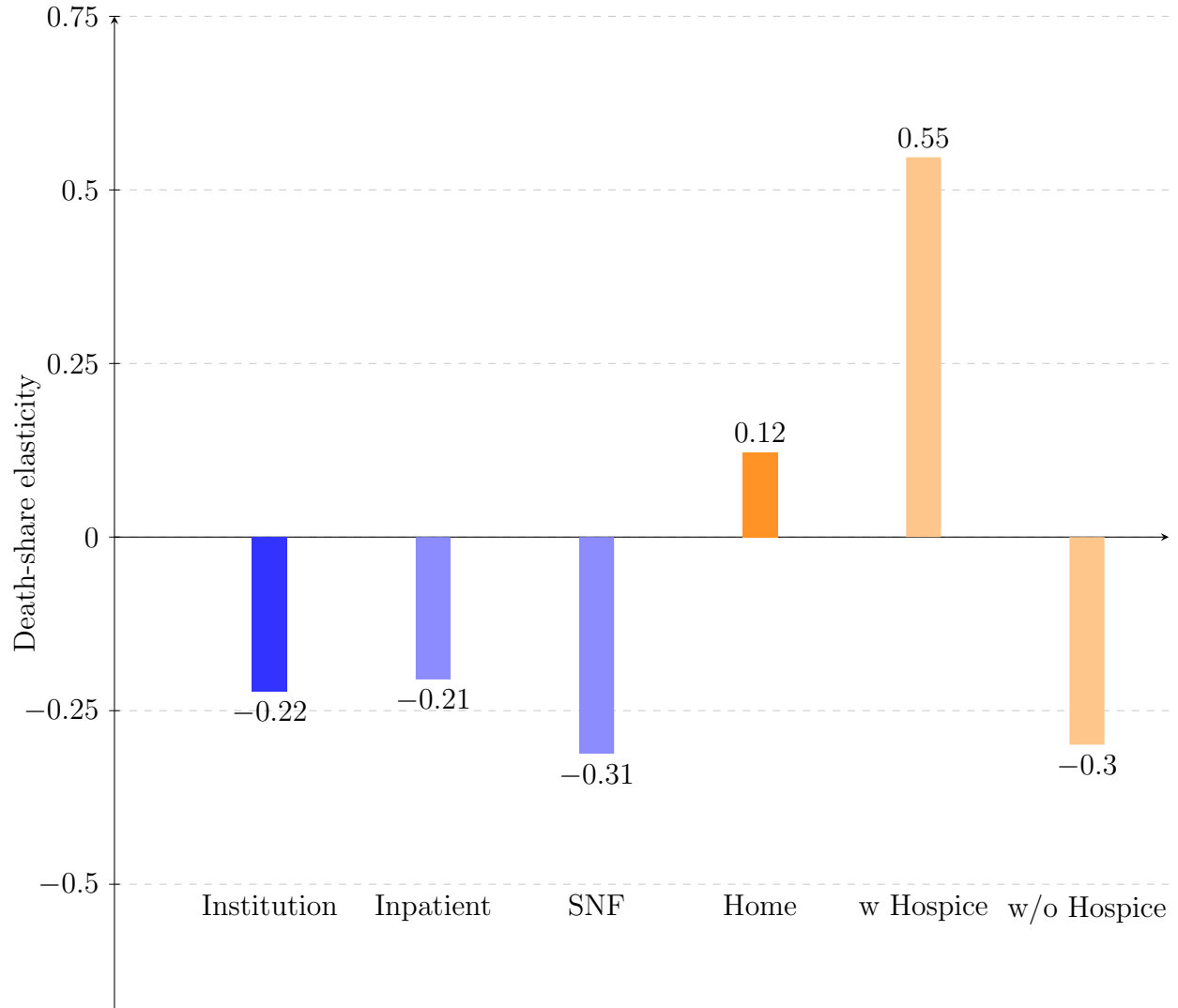
Note: This figure shows estimates from Equation (1) of hospice agency entry on Medicare fee-for-service spending by care setting: inpatient, skilled nursing facility, outpatient, home health, office visits, and prescription drugs (Panels A through F). Estimates are based on a stacked difference-in-differences event study design using 1,518 hospice agency entries, each defining a separate experiment with one treated county and up to two matched control counties selected based on pre-entry Medicare beneficiary levels and trends. The outcome is measured at the county-experiment-quarter level and modeled with county-by-experiment and experiment-by-time fixed effects. Event-time coefficients represent differences between treated and control counties, normalized to zero in the quarter prior to entry ($k = -1$). The red dashed line marks the timing of facility entry. Standard errors are clustered at the experiment level and shown as 95% confidence intervals.

Figure 5. Event Study: Overall Mortality



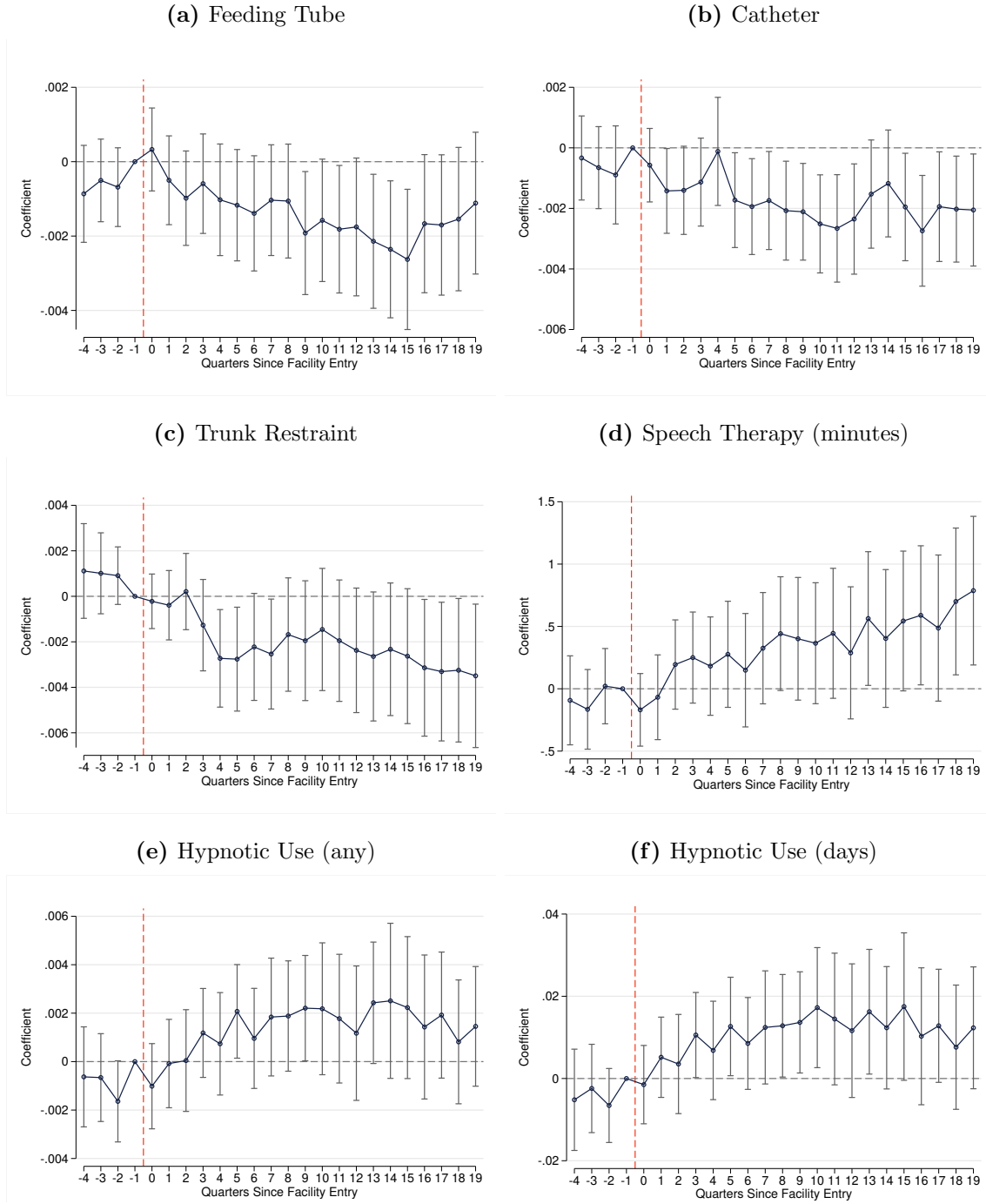
Note: This figure shows estimates from Equation (1) of hospice agency entry on total quarterly deaths per 1,000 beneficiaries. Estimates are based on a stacked difference-in-differences event study design using 1,518 hospice agency entries, each defining a separate experiment with one treated county and up to two matched control counties selected based on pre-entry Medicare beneficiary levels and trends. The outcome is measured at the county-experiment-quarter level and modeled with county-by-experiment and experiment-by-time fixed effects. Event-time coefficients represent differences between treated and control counties, normalized to zero in the quarter prior to entry ($k = -1$). The red dashed line marks the timing of facility entry. Standard errors are clustered at the experiment level and shown as 95% confidence intervals.

Figure 6. Place of Death Elasticities



Note: This figure reports the elasticity of the share of deaths that occur in different settings with respect to the share of beneficiaries using hospice in a calendar quarter. Elasticities are calculated by dividing the percent change in the share of deaths that occur in each setting shown in Table 5 by the percent change in the share of beneficiaries using hospice in a calendar quarter shown in Table 3.

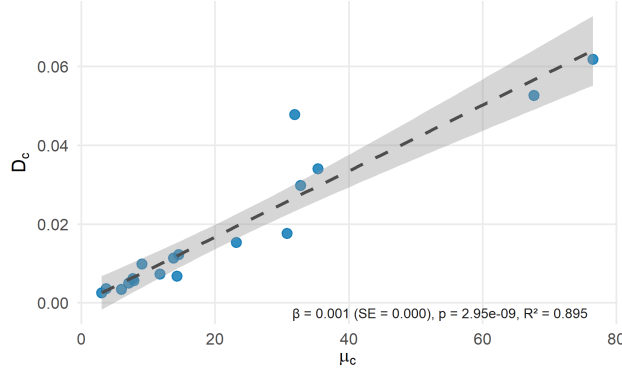
Figure 7. Event Study: Interventions Affecting Patient Comfort



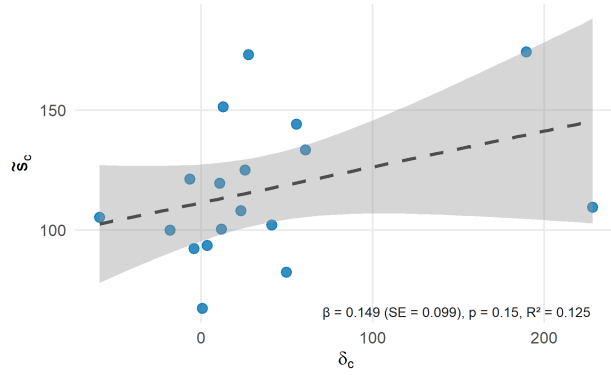
Note: This figure shows estimates from Equation 1 of hospice agency entry on the use of interventions relevant to patient comfort: Panels A-D show feeding tubes, catheters, trunk restraints, and speech therapy for swallowing. Panels E and F show use of hypnotic medications. Estimates are based on a stacked difference-in-differences event study design using 1,518 hospice agency entries, each defining a separate experiment with one treated county and up to two matched control counties selected based on pre-entry Medicare beneficiary levels and trends. The outcome is measured at the county-experiment-quarter level and modeled with county-by-experiment and experiment-by-time fixed effects. Event-time coefficients represent differences between treated and control counties, normalized to zero in the quarter prior to entry ($k = -1$). The red dashed line marks the timing of facility entry. Standard errors are clustered at the experiment level and shown as 95% confidence intervals.

Figure 8. Correlation between Hospice Use and Key Model Parameters

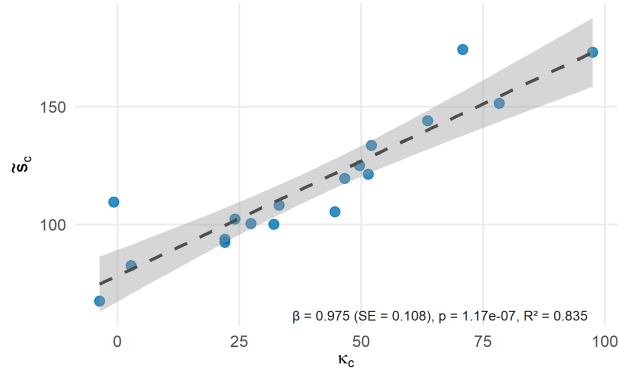
(a) Baseline hospice share D_c vs. mean severity $\mu_{s,c}$.



(b) Severity threshold $\tilde{s}_c$ vs. revenue loss $\delta_{R,c}$.

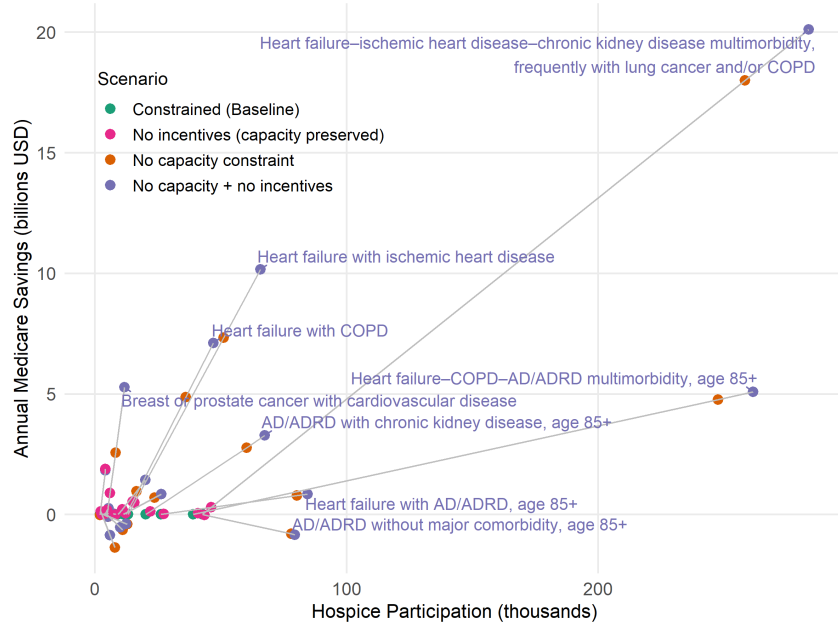


(c) Severity threshold $\tilde{s}_c$ vs. access constraints κ_c .



Note: This figure reports correlations between several key model parameters, across conditions. Panel A shows that conditions with higher average severity have higher baseline hospice participation. Panel B shows that conditions with larger provider revenue losses tend to have higher severity thresholds. Although not statistically significant, this positive relationship is consistent with the model, where financial incentives shift the cutoff for hospice inclusion. Panel C shows that conditions with more binding capacity constraints also exhibit higher severity thresholds, indicating that capacity constraints shift the cutoff for hospice inclusion.

Figure 9. Counterfactual Medicare Savings by Patient Group



Note: Figure shows patient group-specific medicare savings (in billions USD per year) as a function of the quarterly hospice participation rate (in thousands of beneficiaries). Each point corresponds to a unique combination of group and counterfactual scenario. Grey lines connect points within the same group. The slope of each line corresponds to the group-specific value of m_c , which is the annual net Medicare savings from an additional hospice participant (see Section C.5 for methodological details). Patient group descriptions (see Table A14) are shown for groups with the largest savings or hospice participation in the scenario without incentives or capacity constraints.

Table 1. Summary Statistics of Hospice Agencies

Variable	All	Incumbents	Entrants	Entrants in Baseline Sample
Number of Employees	29.94	47.38	18.62	24.68
Registered Nurses	2.19	3.21	1.50	1.54
Licensed Practical Nurses	9.79	16.24	5.53	7.71
Physicians	1.09	1.33	0.97	1.07
Home Health Aides	6.56	9.73	4.36	6.14
Homemakers	0.62	1.01	0.46	0.55
Counselors	1.68	2.20	1.31	1.54
Medical Social Workers	2.34	3.62	1.52	1.90
Other	5.67	10.03	2.96	4.23
Type of Ownership				
Non-Profit	0.30	0.58	0.16	0.22
For-Profit	0.58	0.30	0.72	0.64
Government	0.12	0.12	0.12	0.14
Co-location with Other Health Care Facilities				
Free-Standing Independent Ownership	0.75	0.51	0.86	0.81
Owned by a Home Health Agency	0.19	0.35	0.12	0.15
Owned by a Skilled Nursing Facility	0.02	0.03	0.01	0.02
Owned by a Nursing Facility	0.003	0.007	0.002	0.003
Owned by a Hospital	0.09	0.22	0.03	0.05
Number of Facilities	5,513	2,154	4,362	1,574
Number of County-Quarters	3,802	1,276	3,253	1,518

Notes: This table reports summary statistics for hospice agencies in 2019. The ‘All’ column includes all hospice agencies in the Medicare Provider of Services (POS) file as of 2019. The ‘Incumbents’ column includes all facilities in operation at the beginning of the study period (1999), and the ‘Entrants’ column includes all facilities that started billing Medicare during our study period (1999 to 2019). The ‘Entrants in Baseline Sample’ column includes hospice agency entrants that are in our baseline sample. Co-location is defined based on the listed facility code and address of each facility.

Table 2. Summary Statistics of Medicare Beneficiaries

Variable	Mean	SD	p25	p50	p75
<i>Panel A: Health care spending (per beneficiary-quarter)</i>					
Hospice	60.04	40.35	28.73	53.26	83.15
Inpatient	752.95	167.44	643.72	744.18	854.74
Skilled Nursing	128.99	59.11	87.77	120.99	163.26
Outpatient	254.69	96.69	184.82	239.31	319.05
Home Health	116.18	91.84	63.87	89.36	134.31
Physician Services	469.40	130.24	379.70	468.37	554.59
Total excl. Prescription Drugs and Hospice	1711.74	417.57	1420.56	1702.01	1982.35
Prescription Drugs (Part D)	441.64	80.00	394.09	439.96	482.64
<i>Panel B: Health outcomes</i>					
Mortality (per 1000)	11.56	1.47	10.64	11.61	12.50

Notes: This table reports summary statistics for the main analysis sample, restricting to treatment counties in the year prior to hospice entry and aggregated to the county by entry-year level. All spending variables in Panel A represent average Medicare expenditures per fee-for-service beneficiary per quarter. Total spending excludes prescription drugs and hospice. Mortality in Panel B is reported as the number of deaths per 1,000 beneficiaries per quarter. N = 771 entry events for prescription drugs (Part D) and N = 1518 entry events for all other rows.

Table 3. Change in Health Care Utilization

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
<i>Panel A: Hospice Utilization and Spending</i>			
Any Hospice	0.00947	0.00053*** (0.0001)	5.6%
Hospice Spending	60.0	4.26*** (0.62)	7.1%
<i>Panel B: Other Spending</i>			
Total, Excluding Part D and Hospice	1712	-9.00** (2.90)	-0.53%
Inpatient	753	-5.29** (1.63)	-0.7%
Skilled Nursing	129	-2.12*** (0.59)	-1.6%
Outpatient	255	0.04 (0.79)	0.02%
Home Health	116	-0.57 (1.22)	-0.49%
Physician Services	469	-1.67 (1.20)	-0.36%
Prescription Drugs (Part D)	442	-0.75 (2.46)	-0.17%

Notes: This table reports estimates from Equation (2). Panel A reports estimates of the effect of hospice agency entry on the share of Medicare beneficiaries who have any hospice use in a quarter, and hospice spending per beneficiary-quarter. Panel B reports estimates of the effects of hospice agency entry on quarterly Medicare spending per beneficiary, including total spending (excluding Part D and hospice) and component categories: inpatient, skilled nursing, outpatient, home health, office visits, and prescription drugs (Part D). Column (1) reports the pre-entry sample mean of the outcome in the treatment county. Column (2) reports the estimated average change in the outcome in the five years after hospice agency entry. Column (3) expresses the estimate as a percent change relative to the pre-entry mean. Standard errors, clustered at the experiment level, are reported in parentheses.

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table 4. Change in Intervention Intensity and Comfort-Oriented Services

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
<i>Panel A: Intervention Intensity</i>			
Has Feeding Tube	0.07	-0.0009 (0.0006)	-1.3%
Has Catheter	0.08	-0.0013** (0.0005)	-1.6%
Has Trunk Restraint	0.06	-0.0029*** (0.0011)	-4.8%
<i>Panel B: Comfort-Oriented Services</i>			
Speech Therapy (Swallowing), Minutes	15.6	0.42** (0.17)	2.7%
Any Hypnotic Drug Use	0.06	0.0021*** (0.0008)	3.5%
Days of Hypnotic Drug Use	0.34	0.014*** (0.004)	4.1%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on intervention intensity and comfort-oriented services, measured using MDS assessment data for nursing home residents. Columns are defined as in Table 3.

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table 5. Change in Place of Death

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Institution	0.36	−0.004*** (0.001)	−1.2%
Inpatient Facility	0.29	−0.003** (0.001)	−1.2%
Skilled Nursing Facility	0.07	−0.001 [†] (0.0007)	−1.7%
Home	0.64	0.004*** (0.001)	0.7%
Home with Hospice Care	0.32	0.010*** (0.001)	3.1%
Home without Hospice Care	0.32	−0.005*** (0.001)	−1.7%

Notes: This table reports estimates from Equation (2) of the effect of hospice agency entry on institutional care utilization in the last days of life and on the place of death. To generate this outcome, we partition beneficiary deaths as occurring either in an institution (inpatient versus skilled nursing facility) or home (with or without hospice care). The denominator is the total number of beneficiary deaths in the county-quarter; thus, outcome means for institution and home sum to one. Columns are defined as in Table 3.

[†] $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table 6. Parameter Estimates

Parameter	Mean	SD	Min	Max
<i>Panel (a): Patient Group-specific Parameters</i>				
Mean severity ($E[s]_c$)	22.079	21.083	3.080	76.500
Severity dispersion ($SD[s]_c$)	45.339	72.955	14.559	333.537
Hospice share (D_c)	0.018	0.019	0.003	0.062
Effect of Hospice Entry on Use ($\eta \cdot g_{s,c}$)	0.0011	0.0011	0.0001	0.0034
Revenue loss ($\delta_{R,c}$)	36.474	69.292	-59.081	228.542
Severity threshold ($\tilde{s}_c$)	116.783	29.310	67.306	174.274
Capacity constraint (κ_c)	39.699	27.463	-3.640	97.516
<i>Panel (b): Common Parameters</i>				
Hospice entry effect on threshold ($\hat{\eta}$)	2.784 (0.704)			
Baseline hospice utility ($\hat{\beta}_0$)	-411.486 (146.171)			
Provider altruism ($\hat{\omega}$)	5.811 (1.680)			

Notes: Parameters are estimated for N=18 mutually exclusive groups, constructed following Section C.1. Rows 1 and 2 of Panel (a) report the means and standard deviation of severity s across conditions, where s has units of deaths per 1,000 beneficiaries per quarter. Hospice share D_c represents the share of beneficiaries in hospice each quarter. The effect of hospice entry on use $\eta \cdot g_{s,c}$ corresponds to our first-stage estimate of hospice entry on the quarterly hospice share. Revenue loss $\delta_{R,c}$ is the amount of Medicare spending (excluding hospice spending) offset per quarter for each new beneficiary in hospice in that quarter, measured in thousands of dollars. $\tilde{s}_c$ denotes the severity threshold, above which a beneficiary is referred to hospice. The capacity constraint κ_c denotes the difference between $\tilde{s}_c$ and the unconstrained threshold s_c^* . $\hat{\eta}$ denotes the effect of a hospice entry event on the referral threshold, assumed to be common across groups. $\hat{\beta}_0$ and $\hat{\omega}$ denote the baseline hospice utility and provider altruism parameters. Standard errors are computed using the heteroskedasticity-robust GMM sandwich estimator for $\hat{\eta}$ and using the Bayesian bootstrap with 200 iterations over groups for $\hat{\beta}_0$ and $\hat{\omega}$.

Table 7. Counterfactual Hospice Utilization and Patient Utilities

Outcome	Scenario	Mean	SD	Min	Max
D_c (Hospice Share)	Constrained (Baseline)	0.018	0.019	0.003	0.062
	No incentives	0.020	0.020	0.003	0.067
	No capacity constraint	0.069	0.109	0.002	0.374
	No capacity + no incentives	0.075	0.117	0.005	0.410
Mean Utility, Hospice Users	Constrained (Baseline)	833.5	808.7	207.3	3972.4
	No incentives	755.2	663.1	206.2	3311.1
	No capacity constraint	477.5	534.1	236.3	2505.7
	No capacity + no incentives	395.4	353.6	235.2	1776.0
Mean Utility, All Patients	Constrained (Baseline)	14.0	15.8	1.973	61.3
	No incentives	14.5	16.5	1.998	64.3
	No capacity constraint	22.8	33.9	2.020	130.8
	No capacity + no incentives	22.9	34.0	2.045	131.3
$s_{\tilde{c}}$ (Severity Threshold)	Constrained (Baseline)	116.8	29.3	67.3	174.3
	No incentives	110.5	27.5	67.2	168.3
	No capacity constraint	77.1	11.9	60.6	110.1
	No capacity + no incentives	70.8	0.000	70.8	70.8

Notes: This table reports summary statistics from counterfactual simulations of the hospice entry model under the four policy scenarios described in 5.5. Outcomes include the hospice participation share (D_c), mean utility among hospice users (in \$thousands), mean utility across all patients (in \$thousands), and the estimated severity threshold $\tilde{s}_c$ (units of quarterly mortality per 1,000 beneficiaries). Each cell reports the mean, standard deviation, minimum, and maximum across chronic conditions. All utility calculations exclude $\delta_{R,c}$ to reflect only the nonfinancial components of provider decisionmaking. That is, the hospice utility for a beneficiary with severity s is $\hat{\beta}_0 + \hat{\omega}s$.

Online Appendix

A Construction of MDS Outcomes

Information on care and clinical outcomes for nursing home residents comes from the Minimum Data Set (MDS) ([Centers for Medicare & Medicaid Services, 1999, 2010](#)). The MDS is a screening survey that assesses the health status of residents in Medicare and Medicaid certified nursing homes. Assessments are conducted on admission, discharge, or when there is a significant change in health status as well as annually and quarterly. We only include scheduled assessments: admission, annual, and quarterly. We use MDS data from 1999 through 2019. In October of 2010, the assessment transitioned from MDS 2.0 to MDS 3.0; some assessment measures were updated at this time. In these cases, we harmonize the survey question(s) to allow us to use all of the years of the data.

A.1 Variable definitions and harmonization

Feeding tube use: The assessment records a binary variable for whether the patient has any recorded presence of any type of tube that can deliver nutrition, fluids, or medications directly into the gastrointestinal system (e.g., nasogastric tubes, gastrostomy tubes, PEG tubes, etc.) in the previous 7 days.

Catheter use: The assessment records a binary variable for whether the patient has any use of an indwelling catheter (i.e., a catheter maintained within the bladder for the purpose of continuous drainage) over the past 14 (MDS 2.0) or 7 (MDS 3.0) days.

Trunk restraint: Trunk restraints include any manual method, device, or equipment attached or adjacent to the patient’s body such that the resident cannot easily remove that restricts movement or access to his or her body. The assessment records one of the following values based on frequency of use over the previous 7 days, coded as follows: 0 (trunk restraints not used), 1 (used less than daily), or 2 (used daily).

Therapies: For each of the following types of therapy provided by a licensed provider, the assessment records (1) the number of days on which the patient received 15 minutes or more of therapy out of the previous 7 days and (2) the number of minutes of therapy the patient received. In MDS 3.0, the latter is further disaggregated by mutually exclusive setting (e.g., individual therapy, group therapy, etc.); we take the sum of minutes across therapy settings to harmonize the data across MDS 2.0 and 3.0. We analyzed the following types of therapy:

- **Speech therapy:** services address swallowing disorders (dysphagia) through assessment and interventions such as oral–motor strengthening, compensatory strategies, and therapeutic exercises to improve safety and efficiency of oral intake. Services may also include management of communication and cognitive-communication disorders when present.
- **Respiratory therapy:** services for the assessment, treatment, or monitoring of patients with pulmonary function difficulties including coughing, deep breathing, heated nebulizers, mechanical ventilation, etc.

- Physical therapy: diagnostic and management services to address movement dysfunction such as therapeutic exercise, functional training, assistive and adaptive devices, etc.
- Occupational therapy: interventions intended to address deficits in physical, cognitive, psychosocial, sensory, and other aspects of performance relevant to functioning in everyday life.
- Psychological therapy: treatment of mental and emotional disorders using psychological techniques.

Drug use: For each of hypnotics, antidepressants, anxiolytics, antipsychotics, and diuretics, the assessment records how many out of the previous 7 days the patient received any amount of the relevant drug type. In the MDS 2.0 version, this includes days prior to admission for assessments conducted less than a week after admission. In the MDS 3.0, days prior to admission are excluded. We use these variables as they are originally recorded.

ADLs: For each of the following activities of daily living (ADLs), the assessment records two versions: one to reflect self-performance and another reflecting supported performance. For both measures, higher values reflect worse performance (more assistance required). For self-performance, individuals receive a numeric score of 0 (independent), 1 (with supervision such as encouragement or cueing), 2 (limited assistance including guiding or non-weight-bearing assistance), 3 (extensive assistance including weight-bearing support), or 4 (total dependence) based on the highest level of assistance required 3 or more times over the previous week. For the supported version of the outcomes, the codes are 0 (no help or oversight), 1 (setup help only), 2 (one person assists), or 3 (two or more people assist). There are also codes for the activity occurring fewer than 3 times, which we treat as missing values.

- Bed mobility: performance in moving to or from a lying position, turning side to side and positioning self in bed
- Transfer: performance moving between surfaces or planes
- Walking in room: walking in own room, with or without assistive devices.
- Walking in corridor: walking in corridor of facility, with or without assistive devices
- Locomotion on unit: mobility within facility
- Locomotion off unit: mobility outside of facility
- Eating: how the resident eats and drinks, excluding medication pass but including nourishment by means like tube feeding or IV fluids.
- Toileting: how the resident uses the toilet room, commode, bedpan, or urinal including transfers, self-cleansing, management of catheter, etc.
- Personal hygiene: maintenance of personal hygiene including combing hair, brushing teeth, shaving, etc. excluding bathing and showering.
- Bathing: how the resident takes a full-body bath, shower, or sponge bath and transfers in or out of tub/shower.

Falls: The assessment records information about falls in the previous month and in the previous 2-6 months.¹⁶ For each outcome, we use a binary indicator for having any fall in the specified time window (last 30 days or last 31-180 days). We also use another fall-related outcome: a binary indicator for having a fracture related to a fall in the previous 6 months.

Pressure ulcers: The assessment records whether the patient has any unhealed pressure ulcers. The MDS 2.0 asks about how many unhealed pressure ulcers the patient has at each stage, ranging from stage 1 through stage 4. The MDS 3.0 first asks about whether the patient has any unhealed pressure ulcers and then if so, records how many are at each stage. We take this information and generate two outcomes: a binary indicator for any unhealed pressure ulcers and a numeric outcome indicating the worst (highest) stage.

Incontinence: Bowel and urinary incontinence are reported separately. MDS 2.0 categorizes continence into the following groups: 0 (always continent), 1 (usually continent), 2 (occasionally incontinent), 3 (frequently incontinent), and 4 (always incontinent). The MDS 3.0 uses different categories: 0 (always continent), 1 (occasionally incontinent), 2 (frequently incontinent), and 3 (always incontinent). To harmonize the two versions, we recode the MDS 2.0 categories to pool groups 1 and 2 into group 1, then decrement the categories that were originally encoded 3 and 4 by one. Our harmonized variables are coded as follows: 0 (always continent), 1 (occasionally incontinent), 2 (frequently incontinent), 3 (always incontinent).

Weight changes: The assessment records binary indicators for whether the patient has gained or lost significant amounts of weight, where significant is defined as a change of 5% or more in the last month or 10% or more in the last 6 months.

Pain: Both versions of the MDS ask about frequency of pain and intensity of worst pain. However, the MDS 2.0 asks about pain over the previous 7 days while the MDS 3.0 asks about previous 5 days.

Pain Frequency: MDS 2.0 and 3.0 differ in how they categorize pain frequency. MDS 2.0 uses a 3-point scale (never, less than daily, daily) while MDS 3.0 asks on a 4-point scale (rarely, occasionally, frequently, almost constant), with a separate binary question about any pain at all. We multiply the MDS 2.0 frequency code by 2 to match the scale of the MDS 3.0 version. The final frequency outcome is a 5-point scale with higher values indicating more frequent pain: 0 (never), 1 (rarely), 2 (occasionally/less than daily), 3 (frequently), 4 (daily/almost constant). We also construct a binary indicator for experiencing any pain in the previous 7 days in MDS 2.0 and 5 days in MDS 3.0.

Pain Intensity: For pain intensity, MDS 2.0 uses a 3-point scale (mild, moderate, horrible/excruciating). MDS 3.0 records a 4-point scale (mild, moderate, severe, very severe/horrible) and a version using integer values 0-10. For MDS 3.0, we recast the 0-10 intensity scale to match the distribution of the pain intensity score from MDS 2.0. Our final harmonized scale takes the following values: 1 (mild/[0-4]), 2 (moderate/[5-8]), 3 (severe/[9]), 4 (horrible/excruciating/[10]).

Mood: Mood-related questions differ between assessment versions. First, the MDS 2.0 asks about previous 30 days while MDS 3.0 asks about previous 2 weeks. The MDS 3.0 uses questions from the PHQ9, a validated screening tool (Kroenke, Spitzer and Williams, 2001), where individual questions measure symptom frequency: 0 (≤ 1 day) to 3 (almost every day). We take the sum of the 9 questions comprising the PHQ9 for a score ranging

¹⁶The MDS 3.0 asks about 1 month and 2-6 months. The MDS 2.0 version asks specifically about previous 30 days and previous 31-180 days.

from 0 to 27. The MDS 2.0 also measures symptom frequency (from 0 to 2); however, it predated the creation of the PHQ9 and thus asks different mood-related questions. For MDS 2.0, we generate a numeric score equal to the sum of the 12 mood-related questions reported in Table 3 of [Burrows et al. \(2000\)](#). This score ranges from 0 to 24. Finally, to harmonize between MDS 2.0 and 3.0, we transformed the numeric scores to z-scores within each year. We use patients’ self reported data unless it’s unavailable, in which case we use the staff reports.

Memory: In the MDS 3.0, we use memory interview questions from items C0200, C0300, and C0400, which reflect immediate word recall, temporal orientation (year, month, day), and short term word recall, respectively. If the patient is unable to perform this interview, staff assess whether the resident was able to normally recall the current season, location of their own room, staff names/faces, and that they are in a nursing home. From MDS 2.0, only these staff reports are available (item B3, recall of current season, location of their own room, staff names/faces, and that they are in a nursing home). For both MDS 2.0 and 3.0, we take the sum of scores across individual questions, where higher scores represent better recall. We convert sums to z-scores using the mean and standard deviation within each year and interview type.

Inattention: Cognitive outcomes included inattention (item B5a in MDS 2.0, C1310B in MDS 3.0), disorganized thinking (B5c and C1310C), and level of consciousness/lethargy (B5E and D1310D). These are worded similarly between MDS 2.0 and 3.0. MDS 2.0 encodes each behavior as: 0 (not present), 1 (present but not of recent onset), or 3 (present and a recent change from usual functioning). MDS 3.0 encodes the behavior as: 0 (not present), 1 (continuously present), or 2 (present but fluctuating). We recode both versions to be binary by pooling the second and third categories within each outcome.

IV Medications: Both MDS 2.0 and 3.0 include binary outcomes indicating whether the patient has received any IV medications in the previous 14 days.

B Matching

This section describes our algorithm for matching control and treatment counties. We computed the following distance statistics for the trends and levels in the number of pre-period beneficiaries in the treatment and control counties:

1. Levels: denoting the treated and control group preperiod beneficiary counts as n_t and n_c , respectively, we defined:

$$n_{dist} \equiv \frac{|n_t - n_c|}{\frac{1}{2}(n_t + n_c)},$$

2. Slopes: denoting the growth rate in the number of beneficiaries as g_t and g_c for the treated and control groups respectively, we defined:

$$g_{dist} \equiv \frac{|g_t - g_c|}{\frac{1}{2}(g_t + g_c)},$$

We then required that control counties satisfy the following requirements:

1. The control county cannot share a border with the treatment county

2. The number of beneficiaries in the treatment and control groups must be similar enough, such that $n_{dist} < 1$.

Among the resulting candidate treatment-control county pairs, we standardized n_{dist} and g_{dist} to have mean 0 and standard deviation 1, and then chose the two control counties per treated county with the lowest sum $d \equiv n_{dist} + g_{dist}$, with the constraint that d is less than the 95th percentile.

C Details on Model Estimation

C.1 Grouping Beneficiary Types

Tables A2 and A5 show that hospice entry has heterogeneous effects on hospice use and non-hospice spending across beneficiaries with different chronic conditions. In addition to differences in the degree to which electing hospice offsets other spending, beneficiaries also differ in terms of their underlying mortality rates. Finally, different types of beneficiaries may face heterogeneous access constraints.

To incorporate these dimensions of heterogeneity, we partition beneficiaries into a finite number of groups based on a binary characteristic vector consisting of their age group (below 65, 65-69, 70-74, 75-79, 80-84, and 85+), sex, and vector of chronic conditions (Alzheimer’s disease, ADRD, Chronic Kidney Disease, Ischemic Heart Disease, Congestive Heart Failure, Chronic Obstructive Pulmonary Disease (COPD), Stroke, Breast cancer, Colorectal cancer, Prostate cancer, and Lung cancer). Because there are about 16,000 unique characteristic vectors, we group beneficiaries with similar characteristics. Specifically, we calculate weighted Hamming distances across characteristic vectors (hereafter, “types”). The weights we use come from an OLS regression of average mortality on each binary characteristic. Thus, distances are smaller between types that only differ in a small set of characteristics that are not important for predicting mortality.

In order to ensure sufficient sample sizes for each group, we require that each group has at least 75 million observations at the beneficiary-county-quarter level between 1999 and 2019. We use a greedy merging algorithm. First, we set aside types that already satisfy the sample size threshold. Then, starting with the type with the fewest observations, we merge in the most similar types (those with the smallest weighted Hamming distances) until the sample size threshold is satisfied. Then, we proceed with the next smallest type that remains. This continues until we have formed groups that all satisfy the sample size threshold, or no un-merged types remain.

C.2 Group-level Empirical Moments

For each merged group, we then obtain the empirical moments we use in the estimation procedure described in the following section. These moments include the baseline hospice participation rate (defined as the share of beneficiaries using hospice in a given quarter), the mean mortality rate, and stacked difference-in-differences estimates of the effect of hospice entry on hospice participation, the effect of hospice entry on non-hospice spending (divided by the first-stage effect on hospice participation), and the effect of hospice entry on average mortality rates among hospice users and non-users. While we do not find any causal effects of hospice on mortality, moving beneficiaries onto hospice will change

the mortality composition of both hospice users and non-users. We use these mortality changes when estimating our model.

A sample size threshold of 75 million yields 27 groups. However, we exclude 9 groups with first-stage estimates below 0.0001 (0.01 percentage points). Hospice use is rare for these groups, and excluding them ensures that our estimates of non-hospice spending offsets are not skewed by noisy first-stage estimates in the denominator. Thus, our model estimates are based on the 18 remaining groups. The resulting groups are shown in Appendix Table A14.

C.3 Estimating Patient Severity Distributions

To take our model to the data, we combine the above empirical moments with the parametric assumption that severity s is log normally distributed within each group c :

$$\ln s \sim \mathcal{N}(\mu_c, \sigma_c^2).$$

We assume that severity s represents a beneficiary's mortality rate (per 1000) in a given quarter. Thus, $E[s]_c$ represents the average mortality rate in group c . We let $\tilde{s}_c$ denote the realized severity cutoff for group c , accounting for access constraints. Beneficiaries are admitted to hospice if $s > \tilde{s}_c$. We recover the parameters $(\tilde{s}_c, \mu_c, \sigma_c^2)$ as follows. First, note that for a log-normal distribution, the observed average mortality for group c has the following closed-form relationship with the parameters (μ_c, σ_c^2) :

$$E[s]_c = \exp\left(\mu_c + \frac{1}{2}\sigma_c^2\right). \quad (5)$$

Next, note that the hospice participation share is given by

$$D_c \equiv 1 - F(\tilde{s}_c; \mu_c, \sigma_c^2), \quad (6)$$

where F denotes the CDF of the log-normal severity distribution. Empirically, we take D_c to be the share of beneficiaries with group c with any hospice use in a given quarter during the baseline period prior to hospice agency entry.

We also observe the first-stage change in D_c when a new hospice agency enters the market, which represents the effect of decreasing the access constraint κ_c on the hospice participation share D_c . Denoting the hospice entry event by Z , and given the definition of D_c , $\tilde{s}_c = s_c^* + \kappa_c$, and $\kappa_c > 0$, it is straightforward to show that

$$\frac{dD_c}{dZ} = -\frac{dF(\tilde{s}_c)}{d\tilde{s}_c} \frac{d\tilde{s}_c}{dZ} = f(\tilde{s}_c; \mu_c, \sigma_c^2) \cdot \eta, \quad (7)$$

where $f(\tilde{s}_c; \mu_c, \sigma_c^2)$ is the log-normal density evaluated at $\tilde{s}_c$ and $\eta \equiv -\frac{d\tilde{s}_c}{dZ}$ is the effect of hospice entry on the referral threshold $\tilde{s}_c$, or equivalently, on the access constraint κ_c . We assume that η is constant across groups for tractability.

Note that for any candidate value of η , Equations (5), (6), and (7) can be directly solved for $(\tilde{s}_c, \mu_c, \sigma_c^2)$, because we directly observe D_c , the group-specific mean severity $E[s]_c$, and the first-stage effects $\frac{dD_c}{dZ}$.

To estimate η , we use the intuition that shifting the threshold $\tilde{s}_c$ will change the average mortality of beneficiaries admitted to hospice and those remaining outside hospice. Let

$s'_c \equiv \tilde{s}_c - \eta$ denote the new referral threshold following hospice entry.

Because severity s corresponds to a beneficiary's mortality rate, the change in observed mortality among hospice users following hospice entry reflects the change in the expected severity of individuals above the admission threshold. Specifically, the change in hospice mortality for group c is given by

$$\Delta m_c^{\text{hos}} = E[s \mid s > s'_c] - E[s \mid s > \tilde{s}_c]. \quad (8)$$

Similarly, hospice entry also changes the composition of individuals who remain outside hospice. The change in mortality among non-hospice beneficiaries is therefore

$$\Delta m_c^{\text{non}} = E[s \mid s \leq s'_c] - E[s \mid s \leq \tilde{s}_c]. \quad (9)$$

Under the log-normal severity assumption, the conditional expectations in Equations (8) and (9) have closed-form expressions as functions of $(\tilde{s}_c, \mu_c, \sigma_c^2)$ and the shifted threshold s'_c . Thus, for any candidate value of η , we can compute the implied mortality changes $(\Delta m_c^{\text{hos}}, \Delta m_c^{\text{non}})$ for each group c .

Empirically, we used our stacked difference-in-differences framework to estimate the corresponding reduced-form effects of hospice entry on mortality among hospice and non-hospice beneficiaries, denoted $(\widehat{\Delta m}_c^{\text{hos}}, \widehat{\Delta m}_c^{\text{non}})$. We therefore estimate η by choosing the value that minimizes the discrepancy between the model-implied and empirically observed mortality changes across groups. Specifically, we estimate η using a generalized method of moments estimator that minimizes

$$\hat{\eta} = \arg \min_{\eta} \sum_c \mathbf{g}_c(\eta)' W \mathbf{g}_c(\eta), \quad (10)$$

where

$$\mathbf{g}_c(\eta) = \begin{pmatrix} \Delta m_c^{\text{hos}}(\eta) - \widehat{\Delta m}_c^{\text{hos}} \\ \Delta m_c^{\text{non}}(\eta) - \widehat{\Delta m}_c^{\text{non}} \end{pmatrix}$$

contains the hospice and non-hospice mortality moments for group c , with the same 2×2 weighting matrix W applied to each group.

In practice we implement a two-step GMM estimator in which W is initially set to the identity matrix and then updated using the inverse of the sample variances of the hospice and non-hospice moment conditions across groups. Standard errors are computed using a heteroskedasticity-robust GMM sandwich variance estimator.

Below, we describe how we use these estimates of $(\tilde{s}_c, \mu_c, \sigma_c^2)$ and η in a second estimation step to separately identify the provider's ideal threshold s_c^* and the access constraint κ_c .

C.4 Estimating Provider Utility Parameters

To connect Equation 3 to observables, we impose an affine approximation to the provider's utility of hospice relative to curative care:

$$\Delta U(s) = U_0 + \lambda s, \quad \lambda > 0.$$

Also, we allow the revenue differential ΔR to vary by group, and denote this by δ_c . Plugging these terms into (3) and solving for the value of s that equates the value of hospice to

zero yields the unconstrained cutoff

$$s_c^* = \frac{\delta_c - wU_0}{w\lambda} = \frac{\delta_c - \beta_0}{\omega}, \quad (11)$$

where we define $\beta_0 \equiv wU_0$ and $\omega \equiv w\lambda$. We measure δ_c in units of thousands of dollars per beneficiary in hospice in a given quarter. Also recall that we measure severity s as a given beneficiary's quarterly mortality rate.

Thus, the parameter ω is the value (in thousands of dollars) that providers place on the utility of hospice for a patient with a 0.1 percentage point higher quarterly mortality rate. β_0 is the dollar value of hospice for a hypothetical patient with zero severity.

The realized cutoff is the unconstrained cutoff adjusted by a group-specific access constraint:

$$\tilde{s}_c = \frac{\delta_c - \beta_0}{\omega} + \kappa_c. \quad (12)$$

κ_c captures the tightness of the access constraint (in severity units, i.e., quarterly mortality per 1,000 beneficiaries). For example, a value of $\kappa_c = 10$ means that the mortality threshold needed to qualify for hospice is 1 percentage point higher than the provider's ideal threshold s_c^* .

With the group-specific parameter estimates in hand, we then use a combination of moment inequalities and equalities to recover (β_0, ω) .

The first set of moment inequalities is a lower bound on the value of the threshold $\tilde{s}_c$, which follows from (11) and $\kappa_c \geq 0$:

$$\tilde{s}_c \geq \frac{\delta_c - \beta_0}{\omega}. \quad (13)$$

We require this moment inequality to hold for groups where the access constraint is binding, that is, $\kappa_c > 0$. This moment defines a lower bound on the hospice utility parameters; thus limiting valid parameters to the upper right region of (β_0, ω) -space.

In order to pin down the *level* of κ_c , we normalize $\kappa_c = 0$ for a small subset of groups. For these groups, we require that (13) holds with equality. This assists with achieving point identification, because it defines a line in (β_0, ω) space. In practice, we normalize 3 out of 18 groups; the identity of these groups is chosen by the estimation procedure as those with the smallest differences between $\tilde{s}_c - s_c^*$.

Finally, we impose a 3rd moment inequality that requires the log normal density at the unconstrained threshold s_c^* to exceed the density at the observed (capacity-constrained) threshold $\tilde{s}_c$. This group exploits the fact that baseline hospice use is relatively rare across all groups, such that the shape of the log normal density is declining from s_c^* to $\tilde{s}_c$. This enforces an upper bound on the unconstrained hospice utility.

Identification . The above moments define the following set of moment inequalities or equalities for every group.

$$\begin{aligned} m_{c, \kappa_c \geq 0}^{(1)}(\beta_0, \omega) &= \tilde{s}_c - \frac{\delta_c - \beta_0}{\omega} \geq 0, \\ m_{c, \kappa_c = 0}^{(1)}(\beta_0, \omega) &= \tilde{s}_c - \frac{\delta_c - \beta_0}{\omega} = 0, & \forall c. \\ m_c^{(2)}(\beta_0, \omega) &= f(s_c^*; \mu_c, \sigma_c^2) - f(\tilde{s}_c; \mu_c, \sigma_c^2) \geq 0. \end{aligned} \quad (14)$$

Identification of the utility parameters depends on the intersection of the allowed regions defined by the moments. The moment inequality $m_{c,\kappa_c>0}^{(1)}$ defines an allowed region in the upper right of the (β_0, ω) space while the moment inequality $m_c^{(2)}(\beta_0, \omega)$ allows the lower-left region. In practice, these two moment conditions achieve only set-identification of a diagonal region, where increasing ω can be rationalized by decreasing β_0 . In order to achieve point identification, the moment equality $m_{c,\kappa_c=0}^{(1)}(\beta_0, \omega)$ defines a line in (β_0, ω) space, which is intersected with the above diagonal region. Figure A14 shows the resulting moment violation surface. To obtain standard errors, we employ a Bayesian bootstrap over groups, using 200 bootstrap iterations.

C.5 Computation of Total Hospice Participation and Medicare Savings

We evaluate counterfactual hospice policies by computing group-specific hospice participation and Medicare savings and then aggregating these quantities across patient groups. These results are shown in Figures A15 and 9.

Let D_c^P denote the predicted hospice participation share for group c under a given policy scenario P , and let D_c^0 denote the corresponding participation share under the baseline (constrained) policy. Let $N_{\text{tot}} = 28.7$ million beneficiaries denote the total number of fee-for-service Medicare beneficiaries in 2025¹⁷ and let ρ_c denote the baseline population share of group c , normalized such that $\sum_c \rho_c = 1$ for the original set of 27 groups, with the 18 groups included in our final analysis totaling 0.48. Finally, let m_c denote the net marginal Medicare savings (per beneficiary-year) associated with hospice enrollment for group c , defined as the difference between the expected reduction in non-hospice Medicare spending and the increase in hospice spending.¹⁸

Group-Specific Hospice Participation. Group-specific hospice participation is calculated by scaling the participation share by the group population. The number of hospice participants in group c and policy P is given by

$$H_c^P = N_{\text{tot}} \rho_c D_c^P.$$

Group-Specific Medicare Savings The change in hospice participation relative to baseline for group c and policy P is

$$\Delta D_c^P = D_c^P - D_c^0.$$

The resulting Medicare savings for group c and policy P are calculated as

$$S_c^P = N_{\text{tot}} \rho_c m_c \Delta D_c^P.$$

This expression captures the idea that Medicare savings depend on the number of beneficiaries whose hospice participation changes under the policy and the marginal savings associated with hospice enrollment.

¹⁷See <https://www.kff.org/medicare/a-snapshot-of-sources-of-coverage-among-medicare-beneficiaries/>

¹⁸Annual quantities are obtained by multiplying quarterly quantities by 4. For example, the annual reduction in non-hospice Medicare spending for group c is $4\delta_c$.

Aggregate Hospice Participation Total hospice participation under a given policy scenario is obtained by summing across groups:

$$H_{\text{tot}}^P = \sum_c H_c^P = N_{\text{tot}} \sum_c \rho_c D_c^P.$$

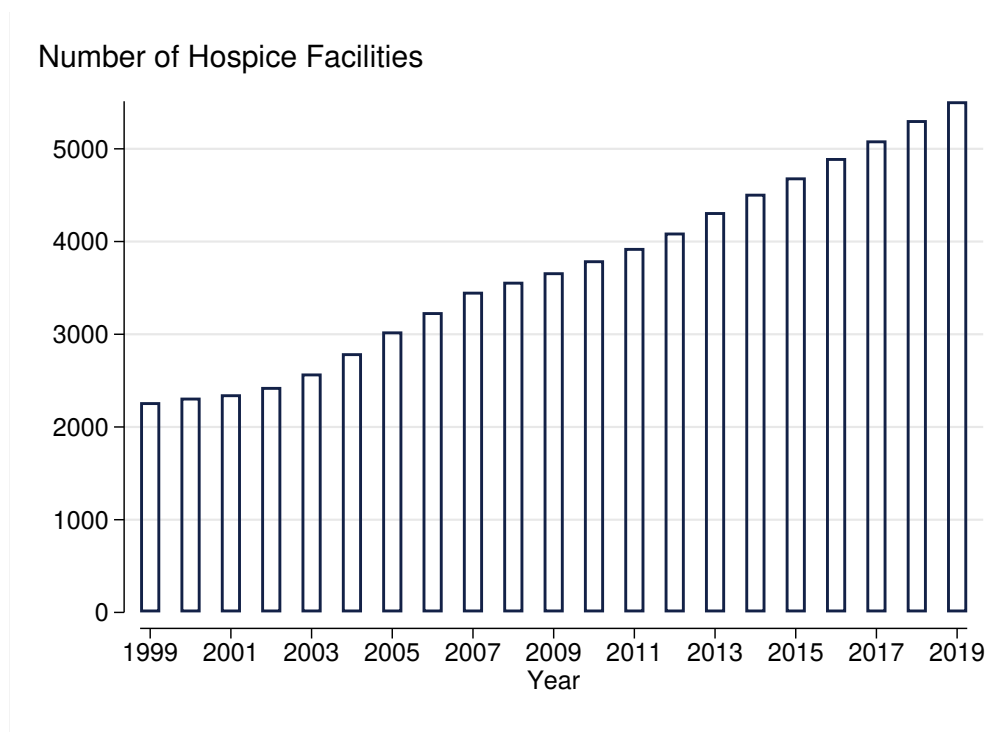
Aggregate Medicare Savings Total Medicare savings, relative to baseline, are calculated by aggregating group-level savings:

$$S_{\text{tot}}^P = \sum_c S_c^P = N_{\text{tot}} \sum_c \rho_c m_c (D_c^P - D_c^0).$$

This expression shows that total savings are a weighted sum of group-specific changes in hospice participation multiplied by the corresponding marginal Medicare savings.

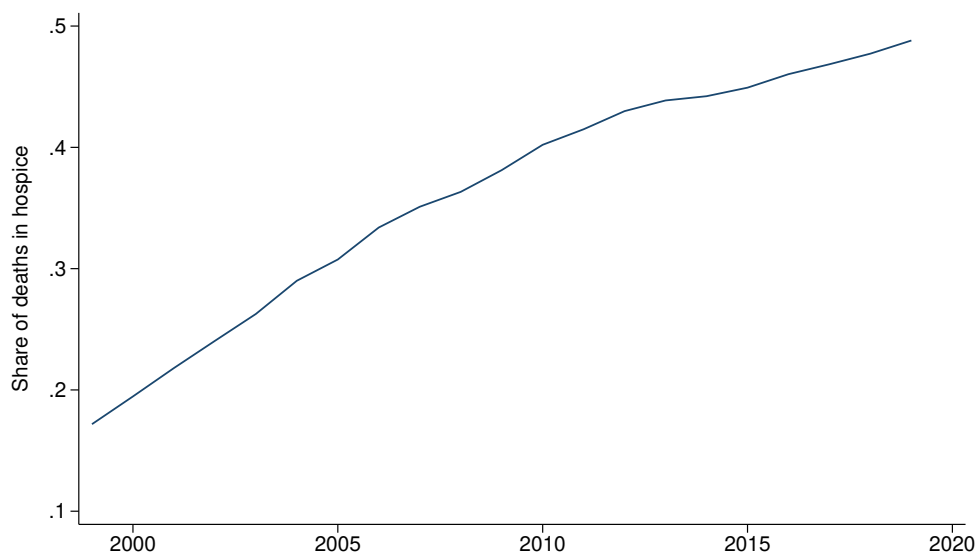
D Appendix Figures and Tables

Figure A1. Number of Hospice Agencies Over Time



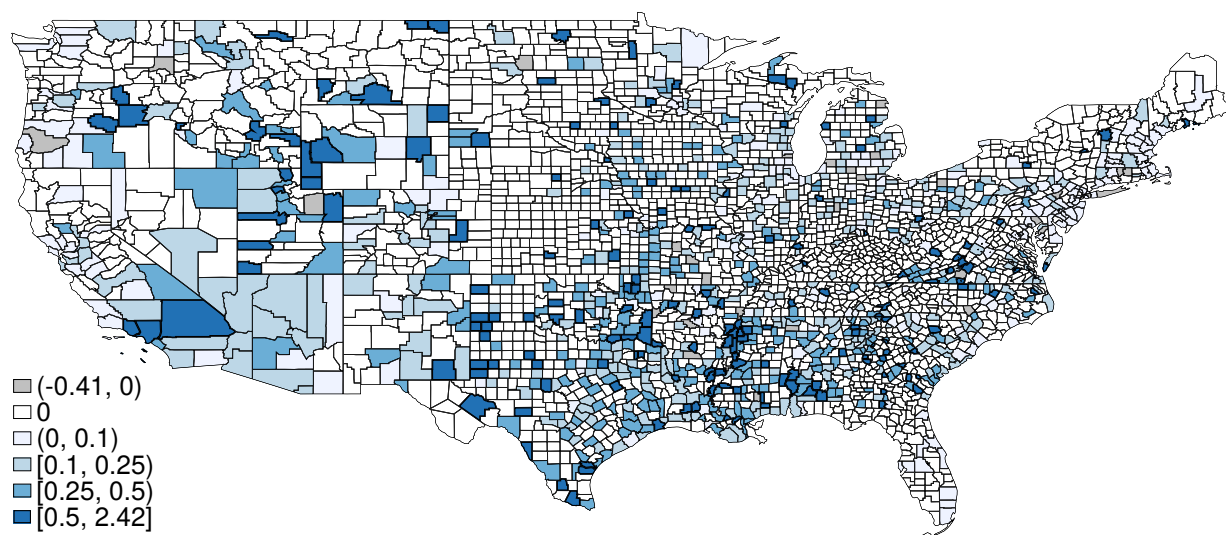
Note: This figure shows the total number of active Medicare-certified hospice providers in the United States from 1999 to 2019. Data are drawn from the CMS Provider of Services (POS) file.

Figure A2. Share of Decedents using Hospice Over Time



Note: This figure shows the share of Medicare beneficiaries who are using hospice at the time of their death. Hospice use at time of death is identified as a beneficiary's date of death coinciding with the end date of a hospice claim.

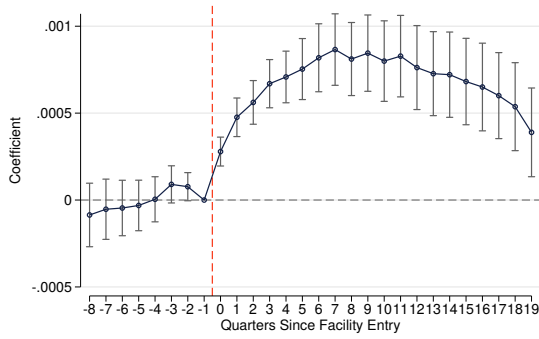
Figure A3. Change in the Number of Hospice Agencies per 1,000 Beneficiaries between 1999 and 2019



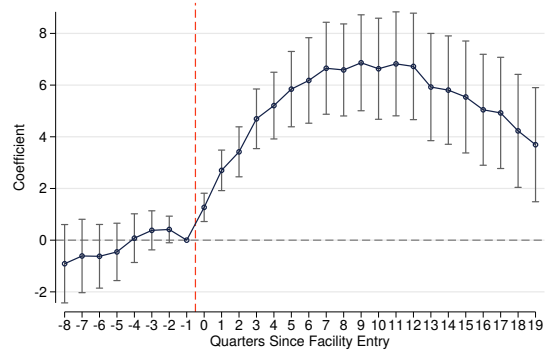
Note: This map shows net change in the number of Medicare-certified hospice agencies per 1,000 beneficiaries by county over the period 1999–2019. Counties are shaded based on the total net increase in agencies per 1,000 beneficiaries during the period. Data shown are from the CMS Provider of Services (POS) file.

Figure A4. Event Study: Hospice Spending (Longer Pre-Period)

(a) Hospice use

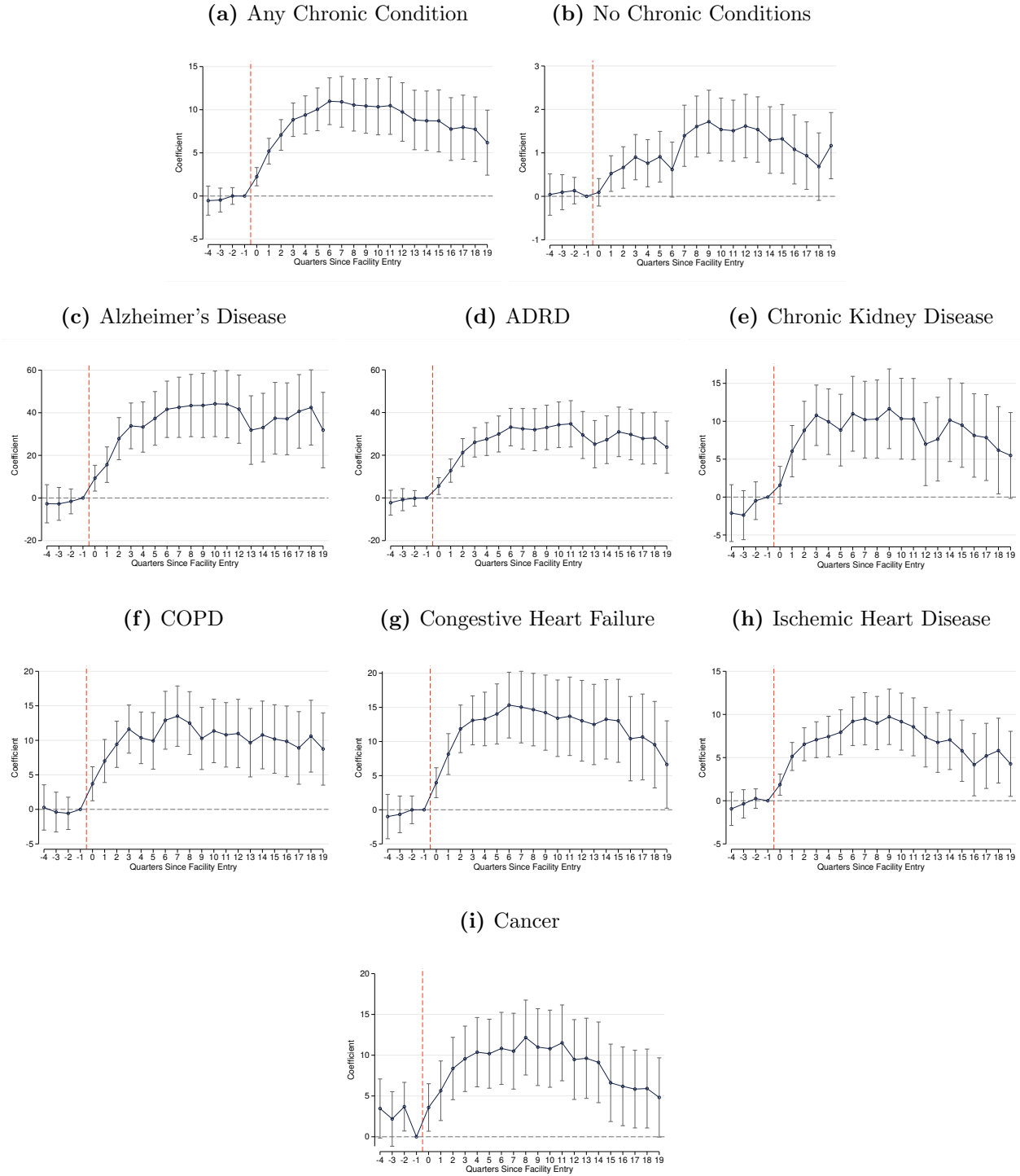


(b) Hospice spending



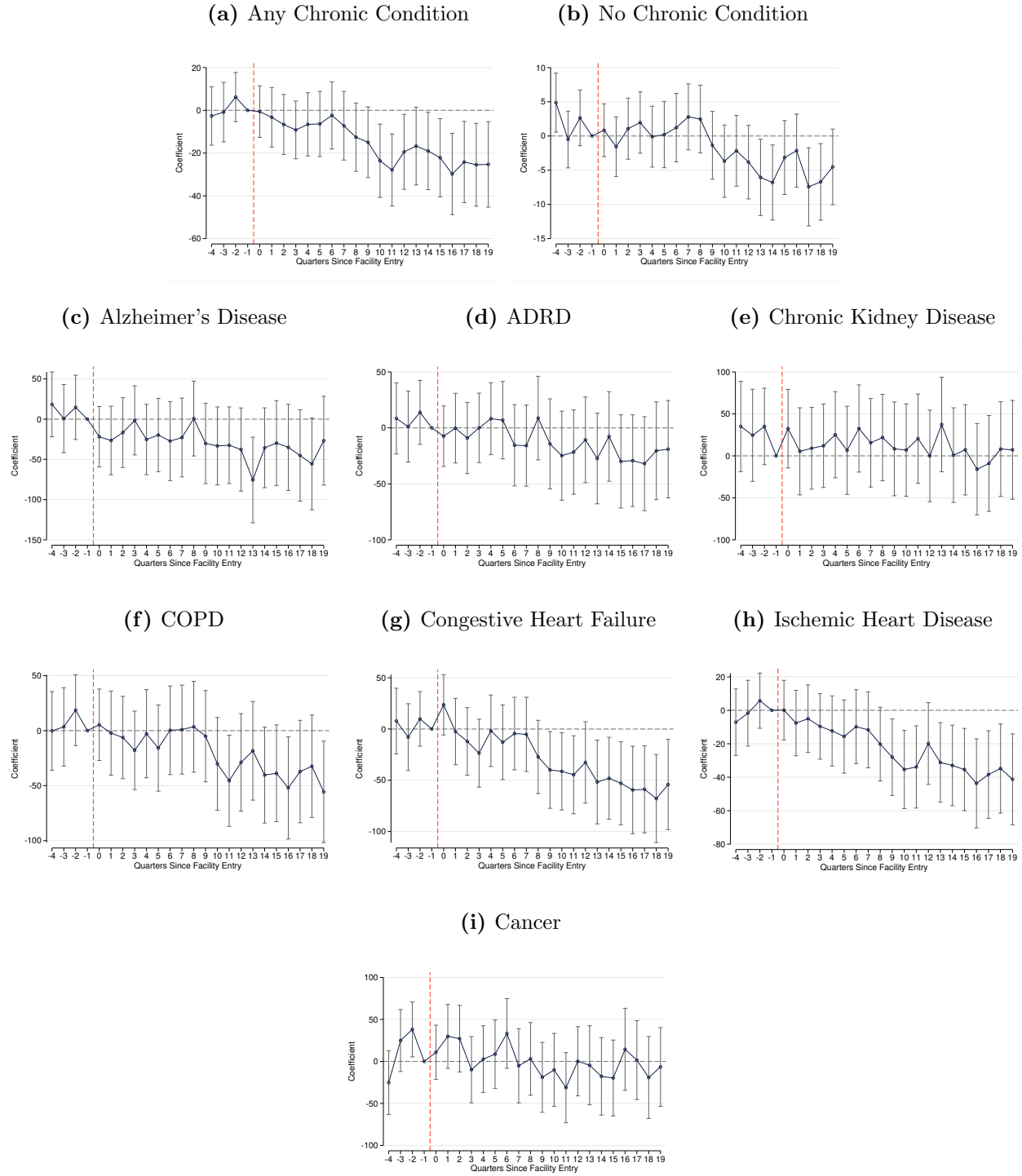
Note: This figure replicates Figure 2 but uses a sample that requires no entry in the 8 quarters prior, instead of 4 quarters as in the baseline. The sample includes 1,147 of the 1,518 experiments analyzed in the baseline analysis.

Figure A5. Event Study: Hospice Spending by Chronic Condition Subgroup



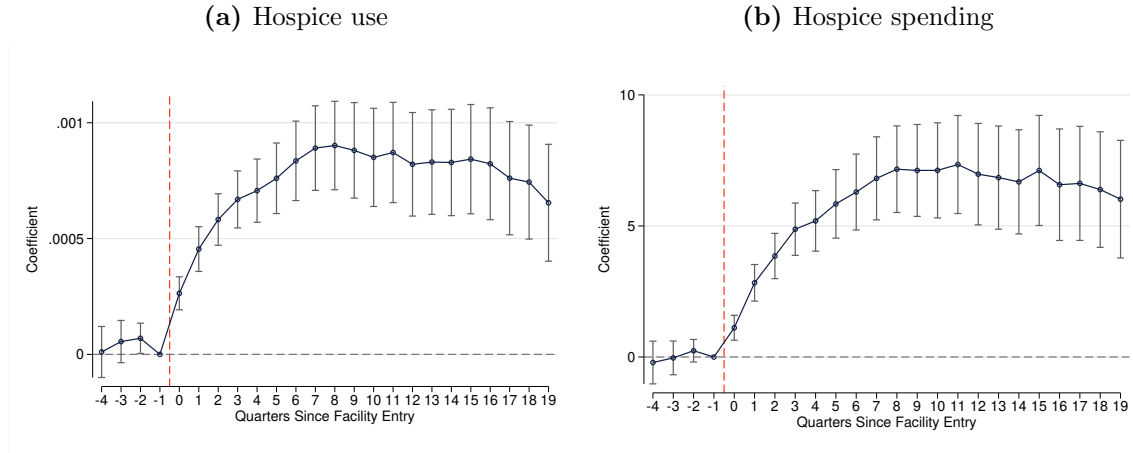
Note: This figure replicates Figure 2 by chronic condition subgroup.

Figure A6. Event Study: Total Spending by Chronic Condition Subgroup



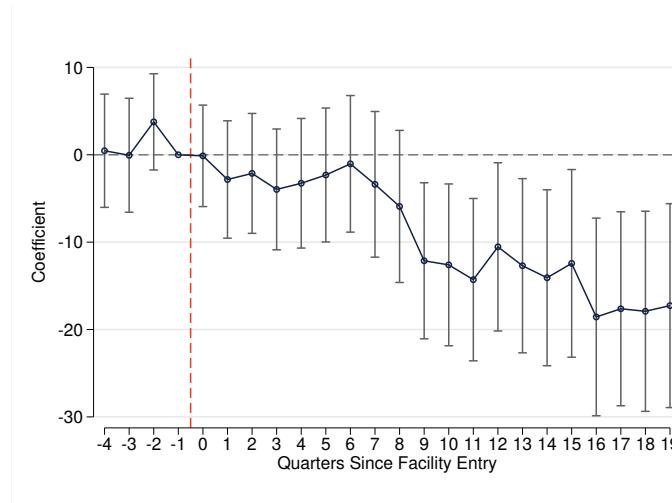
Note: This figure replicates [3](#) by chronic condition subgroup.

Figure A7. Event Study: Hospice Utilization in Cohort-Based Sample



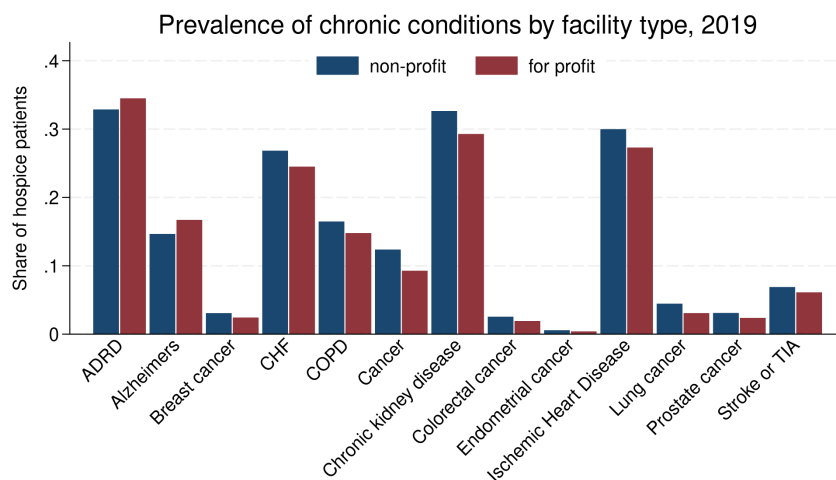
Note: This figure presents estimates of Equation (1) of hospice agency entry on Medicare hospice use per beneficiary per quarter among beneficiaries who were living in the experiment counties for at least one calendar quarter in the pre-entry period. In order for a beneficiary-county-quarter to be included in the sample, a beneficiary must (1) have lived in the county in the pre-treatment period and (2) still be living in that county in a given quarter. Panel A presents the effect on the share of beneficiaries who use any hospice care in a given quarter and Panel B presents the effect of facility entry on hospice spending per beneficiary per quarter.

Figure A8. Event Study: Total Medicare Spending in Cohort-Based Sample, Excluding Part D and Hospice



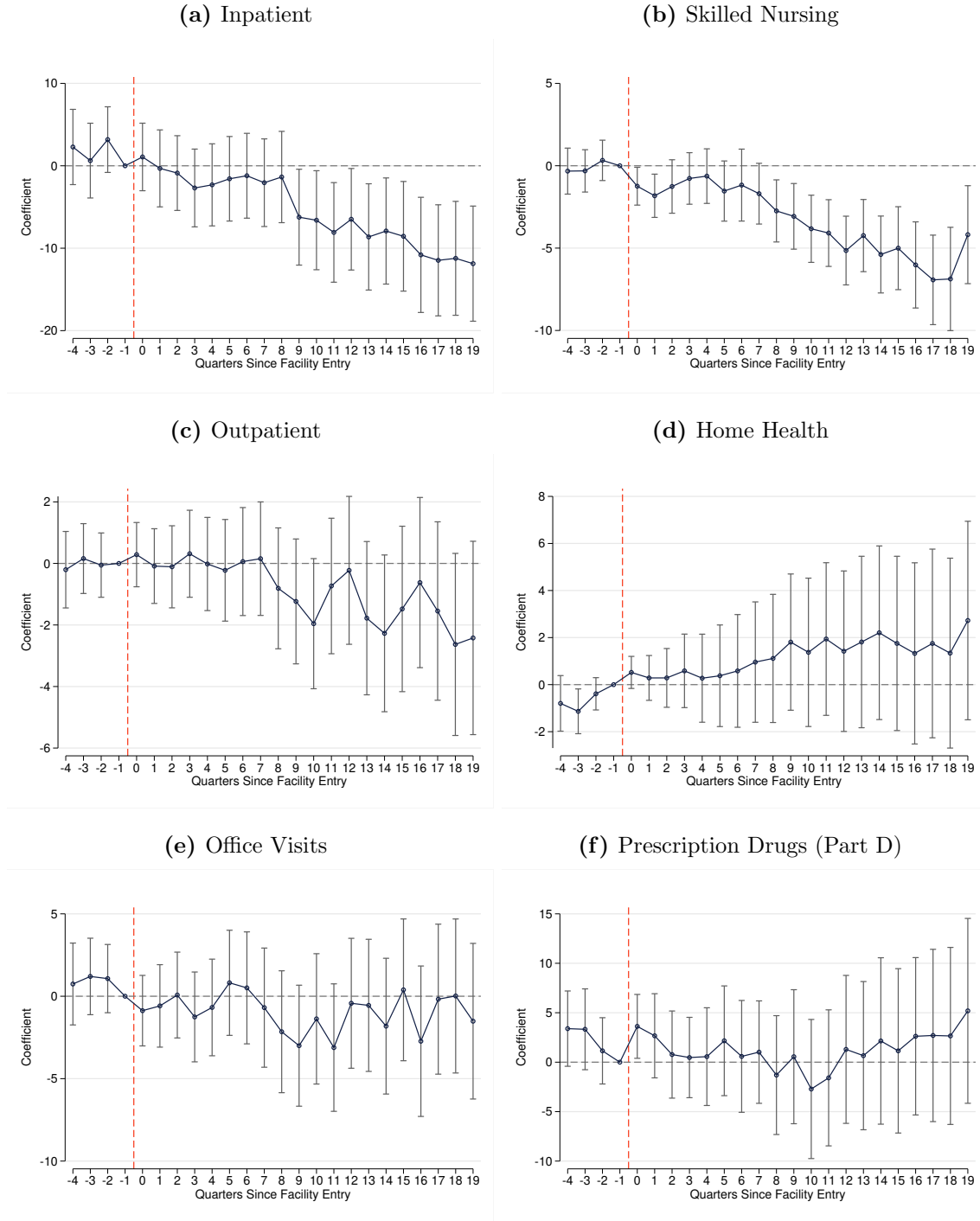
Note: This figure presents estimates of Equation (1) of hospice agency entry on total quarterly Medicare fee-for-service spending per beneficiary, excluding spending on hospice and Part D, among beneficiaries who were living in the experiment counties for at least one calendar quarter in the pre-entry period. In order for a beneficiary-county-quarter to be included in the sample, a beneficiary must (1) have lived in the county in the pre-treatment period and (2) still be living in that county in a given quarter.

Figure A9. Chronic Condition Prevalence by Agency Ownership, 2019



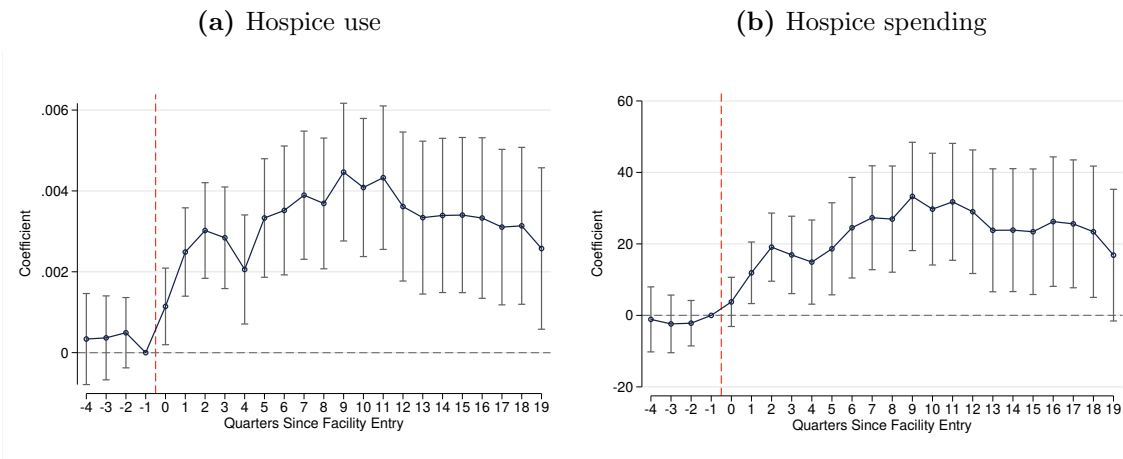
Note: This figure reports the prevalence of major chronic conditions among Medicare hospice patients in 2019 separately for for-profit and non-profit hospice agencies. Each bar shows the share of hospice patients receiving care from agencies of the indicated ownership type who had the corresponding chronic condition.

Figure A10. Event Study: Medicare Spending in Cohort-Based Sample, by Type



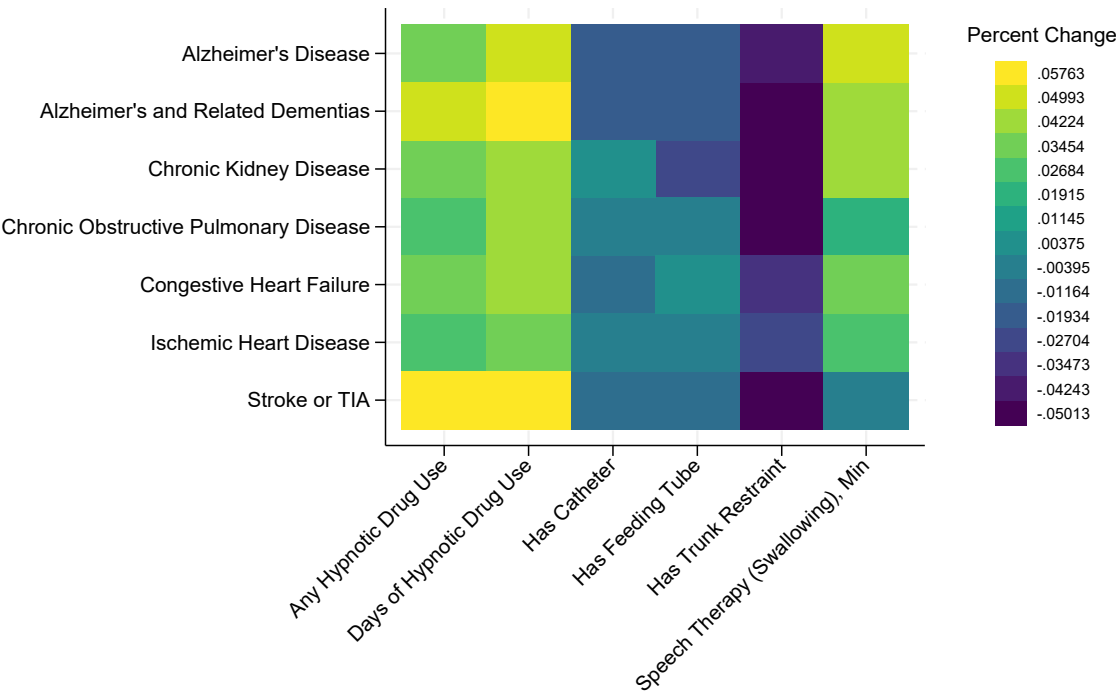
Note: This figure presents estimates of Equation (1) of hospice agency entry on Medicare fee-for-service spending by care setting: Panels A-F show inpatient, skilled nursing facility, outpatient, home health, office visits, prescription drugs, among beneficiaries who were living in the experiment counties for at least one calendar quarter in the pre-entry period. In order for a beneficiary-county-quarter to be included in the sample, a beneficiary must (1) have lived in the county in the pre-treatment period and (2) still be living in that county in a given quarter.

Figure A11. Event Study: Hospice Utilization Among Beneficiaries Living in Nursing Homes.



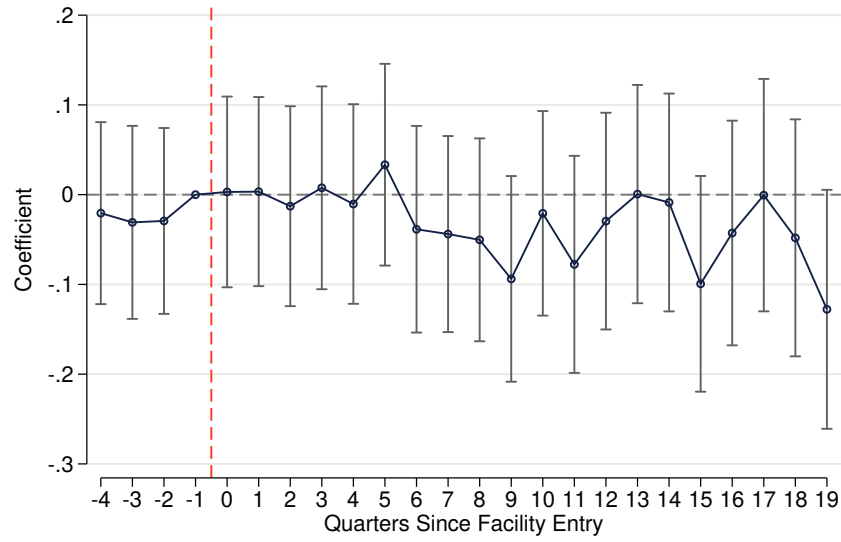
Note: This figure replicates Figure 2 for beneficiaries living in nursing homes.

Figure A12. Change in Interventions and Patient Comfort by Chronic Condition Subgroup



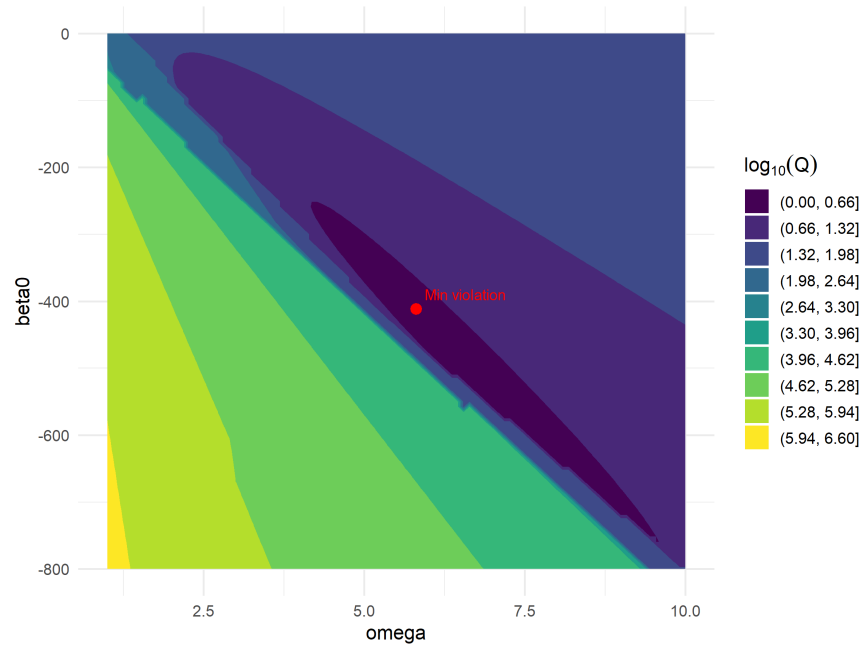
Note: This figure presents estimates of Equation (2) as percent changes relative to baseline means, by chronic condition subgroup and by measures of intervention intensity and comfort-oriented services. The heatmap shows that for each measure (column), the changes appear similar across major chronic conditions.

Figure A13. Event Study: Overall Mortality in Cohort-Based Sample



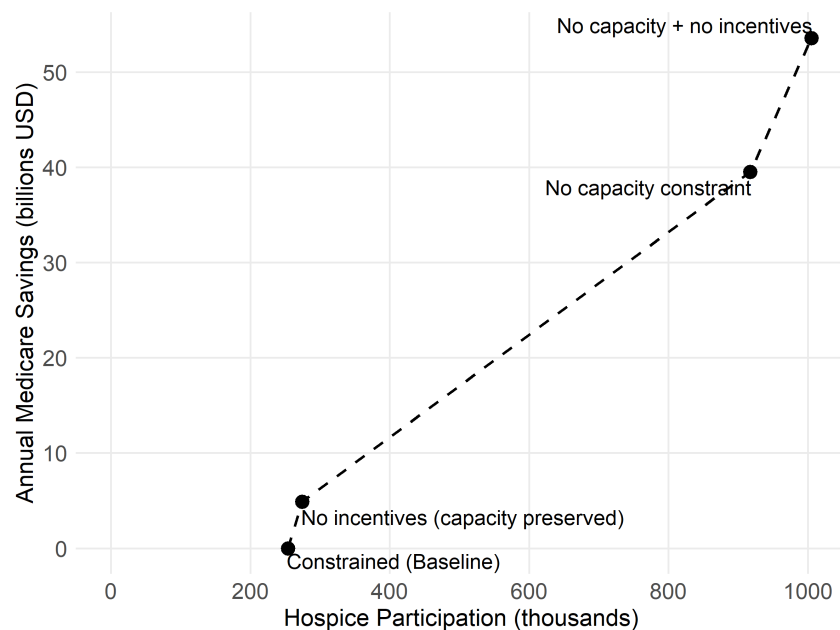
Note: This figure presents estimates of Equation (1) of hospice agency entry on total quarterly deaths per 1,000 beneficiaries for those who were living in the experiment counties for at least one calendar quarter in the pre-entry period. In order for a beneficiary-county-quarter to be included in the sample, a beneficiary must (1) have lived in the county in the pre-treatment period and (2) still be living in that county in a given quarter.

Figure A14. GMM Estimation of Hospice Utility Parameters



Note: Contour plot shows the (log) GMM objective function across values of the hospice utility parameters β_0 and ω . The GMM estimate is shown as the “Min violation” point on the contour plot. See Appendix Section C for more details.

Figure A15. Counterfactual Medicare Savings Relative to Status Quo



Note: Figure shows annual medicare savings (in billions USD per year) as a function of the quarterly hospice participation rate (in thousands of beneficiaries), for each counterfactual policy. See Appendix Section C.5 for details on how savings and participation are computed.

Table A1. County Characteristics

	(1) All Counties	(2) Baseline Counties
Facilities per 1,000 people		
Hospice	0.01 (0.03)	0.01 (0.03)
Skilled Nursing Facility	0.09 (0.09)	0.07 (0.06)
Home Health Agency	0.05 (0.08)	0.05 (0.05)
Demographics		
White (%)	84.37 (16.67)	79.95 (16.34)
Age 65+ (%)	14.74 (4.19)	13.36 (3.71)
Population (1000s)	89.57 (292.42)	190.29 (343.01)
Household Characteristics		
Mean Household Size	2.54 (0.21)	2.57 (0.21)
Median Household Income (\$1000s)	36.34 (9.07)	39.00 (10.69)
Chronic Conditions (per 1000 beneficiaries)		
Alzheimer's Disease	27.77 (8.28)	29.18 (7.65)
Alzheimer's and Related Dementias	67.62 (14.27)	69.04 (13.15)
Chronic Kidney Disease	43.64 (11.82)	44.57 (10.63)
Chronic Obstructive Pulmonary Disease	100.16 (27.54)	97.42 (25.60)
Congestive Heart Failure	159.16 (32.67)	153.69 (31.43)
Ischemic Heart Disease	252.33 (51.25)	249.52 (51.51)
Stroke or TIA	42.85 (11.08)	44.52 (9.94)
Cancer	64.04 (12.16)	64.33 (10.32)
N	3142	891

Notes: This table reports summary statistics on counties as of 2000. Column (1) shows the mean value of the listed variables across all counties in the US with the standard deviation in parentheses. Column (2) shows the same statistics for the set of treated counties in our experiments. Data shown on facilities, demographics, and household characteristics are from the Area Health Resource Files and the 2000 US Census. Chronic condition rates among Medicare are from the Chronic Conditions segment of the Beneficiary Summary File. Chronic condition prevalence is defined using the end of year indicators and is calculated only across counties that had at least 10 beneficiaries.

Table A2. Effects of Hospice Entry on Hospice Spending by Chronic Condition

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Overall	60.0	4.26*** (0.62)	7.1%
Any Chronic Condition	123	8.85*** (1.22)	7.2%
No Chronic Conditions	14.5	1.03*** (0.25)	7.1%
Alzheimer's Disease	477	37.40*** (5.49)	7.8%
Alzheimer's and Related Dementias	377	28.1*** (3.94)	7.4%
Chronic Kidney Disease	152	9.83*** (1.81)	7.8%
Chronic Obstructive Pulmonary Disease	135	10.03*** (1.61)	7.6%
Congestive Heart Failure	189	12.40*** (2.03)	6.6%
Ischemic Heart Disease	104	7.14*** (1.17)	6.9%
Stroke or TIA	149	9.7*** (2.11)	6.5%
Cancer	125	6.27*** (1.40)	5.1%
Breast Cancer	82.8	4.81** (1.75)	5.8%
Colorectal Cancer	172	2.93 (3.36)	1.7%
Prostate Cancer	75.3	5.61** (1.67)	7.4%
Lung Cancer	412	16.3** (5.97)	4.0%

Notes: This table reports estimates from Equation (2) of hospice agency entry on hospice spending per beneficiary-quarter, stratified by chronic condition and cancer type. Columns are defined as in Table 3.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table A3. Effects of Hospice Entry on Hospice Utilization by Pre-entry Hospice Use

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
High Pre-entry Hospice Use			
Hospice Spending	90.8	1.12 (0.95)	1.2%
Total Spending	1884	-6.37 [†] (3.83)	-0.3%
Low Pre-entry Hospice Use			
Hospice Spending	29.3	7.41*** (0.77)	25.3%
Total Spending	1540	-11.64** (4.35)	-0.8%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on hospice and total spending per beneficiary-quarter separately for counties with high and low pre-treatment hospice utilization. An experiment is categorized as having high pre-entry hospice use if the treated county had above median hospice spending per beneficiary-quarter in the pre-treatment period. Pre-treatment hospice spending is defined as the average per beneficiary-quarter amount across quarters -4 through -1 before treatment. The median is taken across all treated counties. Estimates are based on a stacked difference-in-differences design with experiment-by-county and experiment-by-time fixed effects. Columns are defined as in Table 3.

[†] $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table A4. Effects of Hospice Entry on Hospice Utilization by Agency Ownership

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
For-profit Entry			
Hospice Spending	61.9	5.23*** (0.78)	8.5%
Total Spending	1717	-7.25* (3.64)	-0.4%
Non-profit Entry			
Hospice Spending	56.8	2.57* (1.00)	4.5%
Total Spending	1702	-12.06* (4.79)	-0.7%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on hospice and total spending per beneficiary-quarter separately for for-profit and non-profit hospice entrants. Columns are defined as in Table 3. [†] $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table A5. Effects of Hospice Entry on Total Spending Excluding Part D and Hospice, by Chronic Condition

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Overall	1712	-9.00** (2.90)	-0.5%
Any Chronic Condition	3192	-15.80** (5.62)	-0.5%
No Chronic Condition	629	-3.70* (1.51)	-0.6%
Alzheimer's Disease	3709	-38.40** (13.30)	-1.0%
Alzheimer's and Related Dementias	4047	-18.90 [†] (11.00)	-0.5%
Chronic Kidney Disease	6300	-12.00 (16.4)	-0.2%
Chronic Obstructive Pulmonary Disease	4970	-26.50* (11.80)	-0.5%
Congestive Heart Failure	5104	-33.30** (11.30)	-0.7%
Ischemic Heart Disease	3525	-22.50** (7.19)	-0.6%
Stroke or TIA	5939	-47.60** (18.00)	-0.8%
Cancer	3896	-10.00 (13.20)	-0.3%
Breast Cancer	3222	-14.50 (17.20)	-0.5%
Colorectal Cancer	5434	-11.40 (31.20)	-0.2%
Prostate Cancer	3123	-3.05 (16.40)	-0.1%
Lung Cancer	7387	-51.70 (37.90)	-0.7%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on total Medicare non-hospice, non-Part D spending per beneficiary-quarter, stratified by chronic condition. Columns are defined as in Table 3.

[†] $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table A6. Changes in Mortality by Chronic Condition

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
<i>Mortality (per 1000 beneficiaries per quarter)</i>			
Overall Mortality	11.6	-0.017 (0.023)	-0.15%
Any Chronic Condition	23.1	-0.044 (0.051)	-0.19%
No Chronic Conditions	3.22	0.0017 (0.015)	0.05%
Alzheimer's Disease	54.3	-0.33 (0.25)	-0.61%
Alzheimer's and Related Dementias	50.8	-0.22 (0.17)	-0.43%
Chronic Kidney Disease	48.3	-0.11 (0.18)	-0.22%
Chronic Obstructive Pulmonary Disease (COPD)	32.9	-0.030 (0.12)	-0.091%
Congestive Heart Failure	41.2	-0.19 (0.11)	-0.47%
Ischemic Heart Disease	22.6	0.029 (0.064)	0.13%
Stroke or TIA	39.2	-0.33 (0.21)	-0.84%
Cancer	27.9	0.079 (0.13)	0.28%
Breast cancer	16.0	0.07 (0.18)	0.42%
Colorectal cancer	35.5	-0.18 (0.36)	-0.50%
Prostate cancer	16.8	0.21 (0.16)	1.26%
Lung cancer	102.0	0.14 (0.70)	0.14%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on mortality, measured as the number of quarterly deaths per 1,000 beneficiaries in each county, separately by chronic condition. Columns are defined as in Table 3.

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table A7. Change in Share of Days in Institutional Care in the Last 180 Days of Life

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Any Institutional Care	0.13	−0.001** (0.0004)	−1.1%
Inpatient	0.07	−0.0004* (0.0002)	−0.6%
Skilled Nursing	0.06	−0.0009** (0.0003)	−1.6%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on institutional care utilization in the last days of life. For each county-quarter observation, the outcome reported is the share of beneficiaries' last 180 days of life (taking place in that quarter) spent in either inpatient or skilled nursing institutions. Columns are defined as in Table 3.

† $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table A8. Change in Health Care Utilization for Beneficiaries in Nursing Homes

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
<i>Panel A: Hospice Utilization and Spending</i>			
Any Hospice	0.0645	0.00294*** (0.00063)	4.56%
Hospice Spending	471	23.97*** (5.90)	5.1%
<i>Panel B: Other Spending</i>			
Total, Excluding Part D and Hospice	7699	-75.87** (26.40)	-0.99%
Inpatient	3658	-38.17* (16.30)	-1.04%
Skilled Nursing	2900	-40.00** (12.30)	-1.38%
Outpatient	615	0.73 (3.54)	0.12%
Home Health	314	-1.95 (2.18)	-0.62%
Physician Services	231	-2.00 (1.23)	-0.87%
Prescription Drugs (Part D)	111	-2.36* (1.14)	-2.13%

Notes: This table replicates Table 3, restricting to beneficiaries who reside in a nursing home.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table A9. Changes in Physical Health Outcomes Following Hospice Agency Entry

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
<i>Fall and Injury Outcomes</i>			
Fracture in Last 6 Months	0.003	0.0001 (0.0001)	3.2%
Fracture in Last 6 Months (excl. admission)	0.0012	0.000002 (0.00007)	0.15%
Fall in Last 30 Days	0.36	0.0013 (0.0014)	0.36%
Fall in Last 31–180 Days	0.15	0.0012 (0.0012)	0.79%
<i>Pressure Ulcers</i>			
Any Pressure Ulcer	0.05	0.0005 (0.0004)	1.0%
Worst Ulcer Stage (1–4)	0.25	0.0007 (0.0013)	0.28%
<i>Continence Outcomes (0–3 scale)</i>			
Bowel Continence	1.29	0.0002 (0.0030)	0.02%
Urinary Continence	1.56	0.0041 (0.0033)	0.26%
<i>Nutrition Outcomes</i>			
Weight Loss	0.09	0.0001 (0.0006)	0.12%
Weight Gain	0.06	−0.0017* (0.0007)	−2.7%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on adverse outcomes related to safety and physical health. “Fracture in Last 6 Months” indicates the presence of any fall-related fracture. The version “excluding admission” omits initial assessments to isolate changes over time. “Fall in Last 30 Days” and “Fall in Last 31–180 Days” are indicators of recent and subacute fall history. “Any Pressure Ulcer” measures presence of any ulcer, while “Worst Ulcer Stage” ranges from 0 (no ulcer), 1 (least severe), to 4 (most severe). Bowel and urinary continence scores range from 0 (always continent) to 3 (always incontinent). “Weight Loss” and “Weight Gain” are binary indicators of significant change in body weight since the last assessment. Columns are defined as in Table 3.

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table A10. Changes in ADL Support Needs Following Hospice Agency Entry

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
<i>ADL Support Code (0 to 3 scale)</i>			
Bed Mobility	1.68	0.0024 (0.0036)	0.14%
Transfer	1.93	-0.0001 (0.0029)	-0.01%
Walk in Room	1.33	-0.0013 (0.0040)	-0.10%
Walk in Corridor	1.36	0.0017 (0.0038)	0.13%
Locomotion On Unit	1.43	-0.0012 (0.0029)	-0.08%
Locomotion Off Unit	1.51	-0.0004 (0.0028)	-0.03%
Dressing	1.86	0.0030 (0.0020)	0.16%
Eating	1.22	0.0052* (0.0024)	0.43%
Toileting	1.75	-0.0029 (0.0039)	-0.17%
Personal Hygiene	1.73	-0.0046 (0.0034)	-0.27%
Bathing	2.05	-0.0020 (0.0016)	-0.10%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on support needs for activities of daily living (ADLs), measured as the ADL support code (ranging from 0, no help or oversight needed, to 3, requiring assistance from multiple people) from the MDS. Columns are defined as in Table 3.

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Table A11. Change in Therapy Services

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Speech Therapy, Minutes	15.6	0.417** (0.169)	2.67%
Respiratory Therapy, Minutes	9.47	0.295 (0.211)	3.11%
Physical Therapy, Minutes	73.4	-0.565 (0.513)	-0.77%
Occupational Therapy, Minutes	63.9	-0.227 (0.473)	-0.36%
Psychological Therapy, Minutes	0.514	0.0119 (0.0242)	2.32%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on therapy services measured using MDS assessment data for nursing home residents. Therapy minutes reflect the total number of minutes received during the previous seven days from licensed providers. Columns are defined as in Table 3. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A12. Changes in Use of Medications Following Hospice Entry

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Antidepressants (Share)	0.45	-0.0006 (0.0015)	-0.13%
Antidepressants (Days)	3.03	-0.0059 (0.013)	-0.19%
Anxiolytics (Share)	0.17	-0.0001 (0.0013)	-0.08%
Anxiolytics (Days)	1.02	-0.0026 (0.007)	-0.25%
Antipsychotics (Share)	0.24	-0.0004 (0.0013)	-0.17%
Antipsychotics (Days)	1.64	0.0035 (0.0086)	0.21%
Diuretics (Share)	0.36	-0.0019 (0.0013)	-0.53%
Diuretics (Days)	2.55	-0.015 (0.0092)	-0.58%
Hypnotics (Share)	0.06	0.0021*** (0.0008)	3.5%
Hypnotics (Days)	0.34	0.014*** (0.004)	4.1%
IV Medication (Any)	0.0194	0.0005 (0.0004)	2.6%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on use of selected medication classes and interventions, as reported in the Minimum Data Set (MDS). Columns are defined as in Table 3. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A13. Changes in Pain and Mental Health Outcomes Following Hospice Agency Entry

	(1) Outcome Mean	(2) Estimate (SE)	(3) % Change
Pain (Any)	0.40	-0.0008 (0.002)	-0.20%
Pain (Numeric Scale)	0.03	-0.00001 (0.003)	-0.03%
Pain (Frequency)	1.0	-0.001 (0.003)	-0.10%
Mood Score (higher = worse)	0.64	-0.005 (0.007)	-0.78%
Memory Score (higher = better)	0.56	0.002 (0.003)	0.41%
Inattention	0.19	0.003 (0.002)	1.8%

Notes: This table reports estimates from Equation (2) of the effects of hospice agency entry on pain, mental health, and cognitive outcomes, averaging over assessments within each beneficiary-quarter. Columns are defined as in Table 3. None of the estimates in this table are statistically significant at the $p < 0.05$ level.

Table A14. Patient Groups in Model Estimation

Group	Clinical Profile	Quarterly Mortality per 1000
1	Heart failure–ischemic heart disease–chronic kidney disease multimorbidity, frequently with lung cancer and/or COPD	76.5
2	Heart failure-COPD-AD/ADRD multimorbidity, age 85+	67.7
3	Heart failure with AD/ADRD, age 85+	35.4
4	AD/ADRD with chronic kidney disease, age 85+	32.8
5	AD/ADRD without major comorbidity, age 85+	31.9
6	Heart failure with ischemic heart disease	30.8
7	Heart failure with COPD	23.2
8	Chronic kidney disease-COPD-ischemic heart disease multimorbidity	14.6
9	Chronic kidney disease with either COPD or ischemic heart disease	14.3
10	COPD with stroke/TIA	13.8
11	Chronic kidney disease with ischemic heart disease	11.8
12	No major chronic conditions, age 85+	9.1
13	COPD with ischemic heart disease	7.9
14	Breast or prostate cancer with cardiovascular disease	7.7
15	Prostate cancer without major comorbidity	7.1
16	Ischemic heart disease with limited comorbidity	6.0
17	No major chronic conditions, age 65-84	3.7
18	Ischemic heart disease without comorbidity	3.1

Notes: Groups are formed following the algorithm described in Appendix Section C.1. The clinical profiles summarize the dominant conditions describing each group.

Table A15. Total Hospice Participation and Medicare Savings Across Counterfactual Policies

Scenario	Hospice Participation (thousands)	Annual Medicare Savings (billions USD)
Constrained (Baseline)	254	0.00
No incentives (capacity preserved)	274	4.87
No capacity constraint	917	39.51
No capacity + no incentives	1,005	53.58

Notes: Table shows, for each counterfactual scenario, total quarterly hospice participation (in thousands of beneficiaries) and annual Medicare savings (in billions USD) relative to the status quo. See Appendix Section C.5 for details on how savings and participation are computed.